SING's Cross-Model Leakage Ranking Is Indistinguishable from a Null-Energy Ranking

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Abstract

SING (Yadid et al., 2026) ranks 13 ImageNet classifiers by how much class semantics "leak" into the null space of their linear heads, measured by the angular shift (AS) of a CLIP-aligned translation when the null component is removed. We audit this ranking. The fraction of feature energy in the null space, ρ, reproduces SING's published order as well as SING's own score does (Kendall τ=0.82 for both; paired difference 95% CI $[-0.36, 0.29]$), and each keeps a partial association with the order given the other (τ=0.61 both ways). SING's ranking is therefore indistinguishable from a null-energy ranking. Our interventions show that AS has a path from null energy to the score that carries no image-specific content. They cannot, however, separate null energy from real null content as the source of the agreement. Replacing each model's null coordinates with content-free surrogates of the same energy reproduces SING's order (point τ 0.49–0.72 on all 13 models, depending on the surrogate), and so does the excess of real over surrogate AS (τ=0.62). The data cannot apportion the order between the two. Removing an equal share of null energy from every model amounts to measuring AS per unit null energy. With SING's released translators this loses the agreement (τ=$-$0.03 and 0.09 on two model sets; 0.61 on the second set without two models). With retrained ridge translators it keeps it (τ=0.61 and 0.75), at the level our earlier pre-registered rule set to reject the energy reading. Shuffling class means across classes in the null space, which preserves energy exactly, keeps the agreement (τ=0.64), but given ρ the class-aligned part has the stronger partial association (0.81 against 0.22). We read this as leaning toward energy; our pre-registration allows no stronger claim than these raw numbers. In semi-synthetic calibration worlds with known content, no estimator we tested meets all three criteria. SING's "DinoViT" translator matches a supervised timm ViT-L/16 checkpoint, and size-corrected scores change the order but none reverses it with confidence. We also applied new pre-registered energy statistics to four published null-space or residual scores. None of the four was flagged as energy-driven. Post hoc positive controls show that the rule flags SING's IS but not SING's AS or a pure energy score, so this count says nothing about whether the confound is specific to SING. The main analyses were pre-registered over six rounds. Two round-4 addenda were written after the first round-4 results. Post hoc analyses are labelled, and we report every pre-registered outcome. We recommend reporting ρ and an energy-only baseline next to any cross-model AS-type null-space score.

1 Introduction

A linear classifier W on features $f \in \mathbb{R}^m$ ignores every component of f in its null space $\ker W$. SING (Yadid et al., 2026) reads class-relevant content in that null space as "leakage": information the head could have used but did not. To measure it, SING maps features into CLIP space (Radford et al., 2021) with a learned linear translator, removes the null component, and records how far the translated feature turns relative to the class text (the absolute shift, AS). It uses AS and an IS/AS ratio to rank 13 ImageNet models and reports that a model labelled "DinoViT" (Caron et al., 2021) leaks least.

Null spaces matter well beyond SING. They appear in work on out-of-distribution failure (Idnani et al., 2023), outlier and OOD detection (Wang et al., 2022; Cook et al., 2020; Zöngür et al., 2025; Sun et al., 2022;

Ben Ammar et al., 2024), failure prediction (Shin & Kim, 2026), overfitting (Rezaei & Sabokrou, 2023) and information hiding (Li & Short, 2024). Invariance-based adversarial examples show that class information can sit in directions a classifier ignores (Jacobsen et al., 2019). In NLP, nullspace projection and closed-form erasure remove a property from representations (Ravfogel et al., 2020; Elazar et al., 2021; Belrose et al., 2023). Linear maps between a vision model and CLIP, the kind of translator SING relies on, have been studied by Moayeri et al. (2023). Probing work has also shown that readouts need controls (Hewitt & Liang, 2019), and similarity scores are known to be dominated by high-variance directions (Davari et al., 2023).

Across architectures, the null dimension d_{null} and the null energy fraction

$$\rho = \mathbb{E}\|\Pi f\|^2 / \mathbb{E}\|f\|^2, \qquad \Pi = V_n V_n^\top,$$

vary by orders of magnitude (Table 11). Removing more energy should move a translated feature further whether or not that energy is semantic. If so, a cross-model comparison of null "semantics" would be tied to energy. This paper tests that possibility.

Contributions.

1. **Energy-only control** (§3). ρ alone reproduces SING's 13-model order as well as SING's score. The two are statistically indistinguishable predictors, and neither subsumes the other.
2. **Interventions and calibration** (§4). Equal-energy removal, six null-content surrogates, a class-mean permutation and retrained translators. AS has a content-free energy path, but the data cannot apportion SING's order between null energy and null content, and the equal-energy result depends on the translator. In semi-synthetic worlds with known null content, none of eight estimators meets all three calibration criteria. Further scores, including a pre-registered organisation score, and a geometric account of the energy path are in Appendix B.
3. **SING's other claims** (§6, §7). SING's "DinoViT" translator fits a supervised timm ViT-L/16. Size-corrected scores change the order without reversing it with confidence, and several class-level claims do not reproduce on our pipeline.
4. **Energy tests for other scores, with positive controls** (§5). Applied to ViM, ActSub, CORE and conditional $\mathcal{V}$-information, the pre-registered energy rule flags none of them. Post hoc, it flags SING's IS but misses SING's AS and a pure energy score, so we report the count as uninformative and give descriptive statistics instead.
5. **Full reporting**. Six rounds of pre-registration with every outcome listed (Appendix A), including the kills of two score searches and the flaw in the flatness rule behind them (Appendix E).

2 Setup

SING's score. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space, W the model's head and V_n an orthonormal basis of $\ker W$ from the SVD of W (absolute singular-value tolerance 10^{-5} unless stated). For a text direction t, the per-image absolute shift for a removed component r is $|\theta(T(f), t) - \theta(T(f - r), t)|$. AS_{SING} is its mean over images with $r = \Pi f$ (full null removal) and the class text (the normalised mean of SING's six prompts) as t. We use SING's released code (commit `e47b37d`) and all 13 released translators on a fixed 10k-image ImageNet-val subset (10 images per class, all 1000 classes). SING's own model comparison also uses 10k features over the 1000 classes; its 16-class, 2000-image subset is the translator-validation set, which we do not use. For the class-level claims we add 200 images (all 50 validation images of jellyfish, porcupine, Arabian camel and sports car).

Reference order and statistics. SING's order is the 13 IS/AS bars of its Supp. Fig. 12b, which we digitised. A lower ratio means more leakage. Agreement is Kendall τ_b between a leakage score and that order. Every correlation with SING's order in this paper uses this sign convention: positive means that more of the score goes with more leakage (a lower IS/AS ratio). Model-level uncertainty comes from 2000 bootstrap resamples of models (seed 0), with percentile and BCa intervals (Efron, 1987). Paired comparisons

of two scores use the same resamples. With n=13 these intervals are wide and often reach 1, so we read them as coarse. Sets: *all 13*; *E12*, the 12 models with $\rho \geq 0.005$ (all but DenseNet121, whose ρ=1.8×10^{-8}); and *S11*, a screened set declared in round 2 that also drops BiT. The screen removes models with $\rho < 10^{-4}$ or with a retained row-space direction of singular value $\sigma < 10^{-3}$ that holds more than 5% of the feature energy. BiT has such a direction (σ=8.9×10^{-4}, 26% of the energy). ResNet50 has one just under the threshold ($\sigma < 10^{-3}$, 4.4% of the energy): at tolerance 10^{-4} that direction joins the null space, ρ rises from 0.146 to 0.190 and $\mathrm{AS}_{\mathrm{SING}}$ falls from 1.68° to 1.33°.

Pre-registration. The main experiments below were declared in writing before their numbers were computed, over six rounds. Two round-4 addenda (R4B and R4C) were written after the first round-4 results were known, and R4C also after part of R4B and some exploratory geometry had run; each file lists the values known when it was written. The ledger is in Appendix A. Results marked *post hoc* or *exploratory* were added after seeing data. Commit times are set by the committer, so the repository cannot prove the order of events by itself. The hosting service's push history orders only commits that were pushed separately: the round-3 pre-registration was pushed on its own about 68 minutes before its results, the round-4 pre-registrations were pushed before theirs, and the round-6 pre-registration was pushed together with its code and synthetic unit tests before any real-model result.

3 Null energy reproduces SING's order

Reproduction. $\mathrm{AS}_{\mathrm{SING}}$ agrees with SING's order at τ=0.82 (95% CI $[0.56, 1.00]$). The like-for-like comparison, our IS/|AS| ratio against SING's IS/AS ratio, gives τ=0.744 (percentile $[0.33, 1.00]$, BCa $[0.12, 0.94]$). SING's ratios are 0.98–3.35 times ours (Table 1). The gap is largest for DinoViT (3.35×) and EVA02 (2.5×), both small-ρ models. SING's model comparison uses the same number of images and classes as ours, so the gap is not explained by the subset size. Its likely sources are SING's specific images, its concept set or its checkpoint (§6), and we cannot separate them without SING's evaluation lists.

ρ alone reproduces the order. Ranking models by ρ alone gives τ=0.82 $[0.56, 1.00]$, the same as $\mathrm{AS}_{\mathrm{SING}}$ (Fig. 1). On S11 the values are 0.78 for ρ and 0.82 for $\mathrm{AS}_{\mathrm{SING}}$. The paired bootstrap difference $\tau(\rho) - \tau(\mathrm{AS}_{\mathrm{SING}})$ is 0.00 with 95% CI $[-0.36, 0.29]$ (BCa $[-0.37, 0.28]$). Both agreements are far from chance (permutation p=5×10^{-5} for each, the smallest value 20000 permutations can give). ρ and $\mathrm{AS}_{\mathrm{SING}}$ agree with each other at τ=0.69 (Spearman 0.87). Each keeps a partial association with SING's order given the other: partial Kendall τ=0.61 both ways, and partial Spearman 0.63 for $\mathrm{AS}_{\mathrm{SING}}$ given ρ and 0.72 for ρ given $\mathrm{AS}_{\mathrm{SING}}$. The data therefore cannot tell the two predictors apart, and ρ does not make $\mathrm{AS}_{\mathrm{SING}}$ redundant. Null dimension alone is a weak predictor (Spearman +0.46 with the leakage order, p=0.12): DeiT3-L (Touvron et al., 2022) has d_{null}=24 yet the worst ratio. A NuSA-style null norm, $\mathbb{E}_i\|\Pi f_i\|/\|f_i\|$, in the spirit of Cook et al. (2020), is an energy proxy rather than a second leakage score: it orders the models almost as ρ does (τ=0.97) and agrees with SING at τ=0.85.

Within a model, AS grows with removed energy. On ResNet50, ResNeXt101, BiT-R50x1 (Kolesnikov et al., 2020) and DINOv2-L (Oquab et al., 2024), all with d_{null}=1048, removing random d-dimensional subspaces of the null space with $d \in \{24, \ldots, 1048\}$ raises AS about tenfold from d=24 to d=1048 on three of four models (Table 19, Fig. 3 in Appendix D). For random subspaces, removed energy is close to linear in d, so this experiment cannot separate energy from dimension. A second experiment mixes high- and low-variance null PCs to vary energy at fixed d and dimension at fixed energy. It fits $\log|\mathrm{AS}| = a\log\rho + b\log d + c_{\mathrm{model}}$ over 76 cells on five models with a=0.431 (cell bootstrap $[0.388, 0.485]$, model-clustered bootstrap $[0.382, 0.464]$, leave-one-model-out 0.408–0.452; the clustered CI rests on five clusters, one with a single cell, so it is not more reliable for being narrower) and a dimension term b=0.010 whose CI covers 0 (Appendix D). A first-order calculation gives a=0.5: if the translator maps the removed component linearly and the residual is small, the angle changes in proportion to $\|\Pi f\| \propto \sqrt{\rho}$. The pre-registered flatness part of this experiment failed (max/min across d at fixed ρ up to 2.29 against a 1.5 bar), so we claim only $a \approx 0.4$–0.5.

Model	ρ	ours IS/|AS|	SING IS/AS	SING/ours
VGG16	0.350	17.75	24.26	1.37
VGG19	0.339	15.56	19.89	1.28
DenseNet121	1.8×10^{-8}	41.48	70.71	1.70
ResNet50	0.146	17.40	24.81	1.43
"DinoViT"	0.005	20.95	70.16	3.35
EffNet-B0	0.130	22.15	36.28	1.64
BiT-R50x1	0.046	24.93	47.21	1.89
ResNeXt101	0.451	12.60	13.88	1.10
ConvNeXt-B	0.013	32.86	37.38	1.14
Swin-L	0.274	21.21	29.18	1.38
DINOv2-L	0.322	15.49	16.61	1.07
DeiT3-L	0.348	9.73	9.51	0.98
EVA02-L	0.009	21.64	54.32	2.51

Table 1: Like-for-like reproduction. Our IS/|AS| on the 10k subset against SING's IS/AS (digitised Supp. Fig. 12b). $\tau(\text{ours}, \text{SING})$=0.744 [0.33, 1.00].

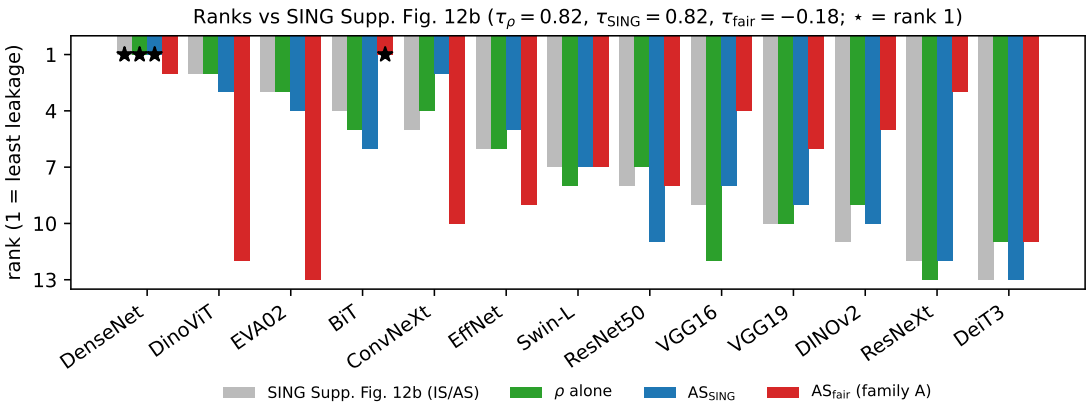


Figure 1: Ranks (1 = least leakage, ⋆ marks rank 1) under SING Supp. Fig. 12b, ρ alone, AS_{SING} and the size-corrected AS_{fair} (family A, §7).

BiT is the exception. Its full-null AS collapses (r=0.06) while its partial subspaces give large AS (9–13°), because the translator's null blocks nearly cancel (top singular value of AV_n 284, against 5.6 for ResNet50). This is also why BiT fails the screen.

4 Energy or content? Interventions

Agreement between ρ and SING's order does not show that energy drives AS across models. Energy could instead *mediate* genuine leakage, if models with more null energy also hold more class content there. We ran four interventions in round 3 and further tests in round 4, pre-registered unless marked post hoc or exploratory (Tables 2 and 8, Fig. 2). None of them separates the two readings. They do show that AS has a content-free path from null energy to the score. They also show that the data cannot apportion SING's order between null energy and null content. We read a τ of 0.6 or more as tracking and $|\tau| < 0.3$ as not reproducing the order; with 13 models, most CIs are wide, so we treat point estimates between the bars as inconclusive. In round 3 we applied the bars unevenly. $\mathrm{AS}_{\mathrm{surr}}$ was read on its point $\tau \geq 0.6$, but X needed its whole CI above 0.6.

4.1 Equal null energy (released translators)

Design. In *A1* we remove $\beta\Pi f$ with $\beta = \sqrt{\rho_0/\rho}$, ρ_0=0.005, so every model loses the same fraction of its feature energy, taken from its own null space in its own proportions. *A2* removes the same energy from the top-variance null PCs, and *A3* removes it from a random rotation of the null basis. *A4* matches energy in CLIP space instead: with $\kappa = \mathbb{E}\|A\Pi f\|^2/\mathbb{E}\|Tf\|^2$, we remove $\beta'\Pi f$ with $\beta'^2\kappa = \kappa_0$=0.0023. The pre-registered outcome was ENERGY-ACCOUNTS if $\tau(\mathrm{A1}) < 0.3$ and the paired CI of $\tau(\mathrm{AS}_{\mathrm{SING}}) - \tau(\mathrm{A1})$ is above 0 on both E12 and S11, and a kill of the energy reading if $\tau(\mathrm{A1}) \geq 0.6$ on both.

What A1 measures. For a small removal, the translated shift is close to linear in the removed component, so A1 $\approx \beta\,\mathrm{AS}_{\mathrm{SING}} = \mathrm{AS}_{\mathrm{SING}}\sqrt{\rho_0/\rho}$: A1 is, to first order, AS per unit null amplitude. The data agree: A1 and $\mathrm{AS}_{\mathrm{SING}}/\sqrt{\rho}$ give nearly the same order (τ=0.79 on E12 and 0.82 on S11; Spearman 0.92 and 0.93), and A1 and $\mathrm{AS}_{\mathrm{SING}}/\rho^{\hat{a}}$ give τ=0.73 and 0.75. The first-order ratio $\mathrm{A1}/(\beta\mathrm{AS}_{\mathrm{SING}})$ runs from 0.36 (DeiT3-L) to 1.00 (DinoViT). The question A1 answers is therefore whether AS *per unit null energy* tracks SING's order. Whether energy causes AS is a separate question.

Result. With the released translators, AS per unit energy does not track SING: τ=-0.03 $[-0.54, 0.61]$ on E12 and 0.09 $[-0.45, 0.78]$ on S11. The paired drop from $\mathrm{AS}_{\mathrm{SING}}$ has CI $[0.23, 1.39]$ (BCa $[0.25, 1.41]$) on E12 and $[0.08, 1.30]$ on S11, which meets the pre-registered ENERGY-ACCOUNTS criterion. Taking the energy from top PCs (0.03) or random directions (0.03) gives the same picture. The result rests on few models. Delete-one jackknife values of τ(A1) span $[-0.16, 0.13]$ on E12 and $[-0.02, 0.29]$ on S11, but dropping DinoViT and EVA02 together gives 0.38 on E12 and 0.61 on S11. The S11 value reaches the level (≥ 0.6) that our round-3 kill rule set for the energy reading, and no other of the 55 leave-two-out subsets of S11 does. Matching CLIP-space energy instead (A4: -0.15 and -0.20) gives the same result, but only with the released translators. With our ridge translators A4 gives 0.52 and 0.49 (§4.4). We therefore do not use A1 or A4 as evidence about the cause of the order. The A1 scores still differ across models by a factor of six (Table 11), so equalising energy changes their order without making them equal.

Translator gain. CLIP-space energy κ tracks SING's order slightly less well than ρ (τ=0.67 on E12). The paired difference $\tau(\kappa) - \tau(\rho)$ has CI $[-0.41, 0.28]$, so we keep ρ as the named variable. The translator's null gain $\gamma = \kappa/\rho$ does not track SING (τ=-0.06). Differences in translator gain γ therefore do not explain the order, although gain does shape the A1 values themselves (§4.4).

4.2 Leak-free null content

Design. We replace each image's null coordinates $c_i = V_n^\top f_i$ with a draw $\tilde{c}_i \sim \mathcal{N}(\mu, \Sigma)$, where μ and Σ are the mean and covariance of c over the 10k images (5 draws). The row-space part, and so every logit, is unchanged. The surrogate carries no image-specific and thus no class-specific content. On ResNet50, ResNeXt101 and DINOv2-L we scale the surrogate to null energy fractions $\rho_t \in \{0.01, 0.05, 0.2, 0.45\}$. We then optionally plant a concept direction with strength $\alpha \in \{0, 0.1, 0.3\}$, using the generator of our holdout test (Appendix E; concepts water, green, thin, knitted). The pre-registered test required that, at α=0, AS rise with ρ_t with Spearman $\geq$0.8 and that the log-log slope CI exclude 0, on all three models.

Score	all 13	E12	S11
ρ (null energy fraction)	0.82 [0.56, 1.00]	0.79 [0.49, 0.97]	0.78 [0.44, 1.00]
$\mathrm{AS}_{\mathrm{SING}}$ (released translators)	0.82 [0.56, 1.00]	0.79 [0.51, 1.00]	0.82 [0.51, 1.00]
κ (CLIP-space null energy)	0.72 [0.44, 0.95]	0.67 [0.37, 0.93]	0.67 [0.35, 0.96]
NuSA-style null norm	0.85 [0.64, 1.00]	0.82 [0.57, 0.97]	0.82 [0.53, 1.00]
$\gamma = \kappa/\rho$ (translator null gain)	−0.10 [−0.56, 0.43]	−0.06 [−0.59, 0.54]	0.05 [−0.51, 0.74]
Leak-free surrogate $\mathrm{AS}_{\mathrm{surr}}$	0.69 [0.39, 0.92]	–	0.64 [0.27, 0.92]
Equal-energy interventions (released translators)			
A1: remove $\beta\Pi f$, $\beta^2\rho = 0.005$	–	−0.03 [−0.54, 0.61]	0.09 [−0.45, 0.78]
A1′: same, $\rho_0 = 0.001$	–	−0.03 [−0.54, 0.61]	0.09 [−0.45, 0.78]
A2: top null PCs, energy ρ_0	–	0.03 [−0.54, 0.54]	0.13 [−0.43, 0.67]
A3: random null rotation, energy ρ_0	–	0.03 [−0.47, 0.67]	0.16 [−0.50, 0.84]
A4: matched CLIP-space energy κ_0	–	−0.15 [−0.68, 0.44]	−0.20 [−0.74, 0.46]
Retrained ridge translators (test half)			
$\mathrm{AS}_{\mathrm{SING}}$	0.85 [0.61, 1.00]	0.82 [0.57, 1.00]	0.82 [0.51, 1.00]
A1	–	0.61 [0.15, 0.94]	0.75 [0.32, 1.00]
A1, λ on extended grid†	0.49 [−0.10, 0.91]	0.39 [−0.21, 0.86]	0.53 [−0.06, 1.00]

Table 2: Kendall τ against SING's order (percentile 95% CI, 2000 model resamples). E12 drops DenseNet ($\rho < \rho_0$); S11 also drops BiT. A1′ gives the same order as A1 because, for each model, AS is monotone in the removed fraction. †Post hoc: ridge penalty chosen on an extended grid $10^{3..6}$, since the pre-registered grid put every model at its upper edge.

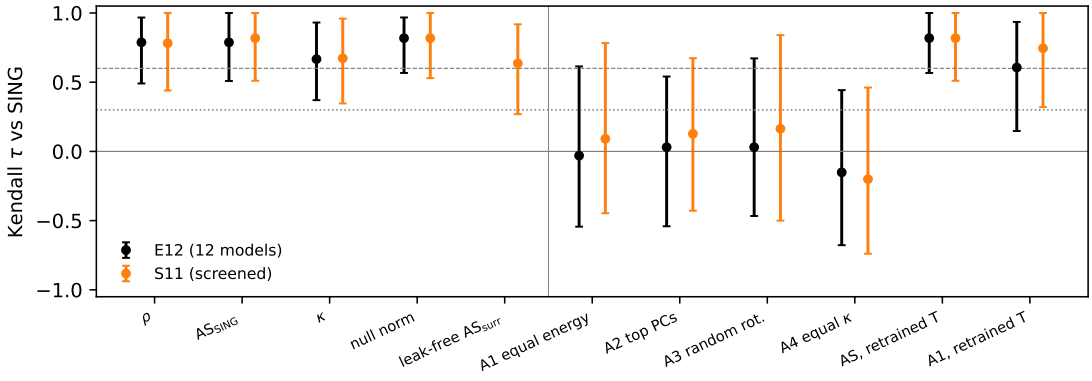


Figure 2: Agreement with SING's order under the energy controls and interventions of §4 (95% percentile CIs). Dotted line: τ=0.3; dashed: τ=0.6, the pre-registered bars.

Result. The test passes on all three models (Spearman 1; slopes 0.38–0.69, all CIs far from 0; Table 12 in Appendix C). AS therefore grows with null energy that carries no image-specific content, which gives it a content-free path from null energy to the score.

Planted content. Planted content raises AS at fixed energy, by ×1.04–1.09 at α=0.1 and ×1.35–1.51 at α=0.3 (ρ_t=0.2), and by ×1.14–1.58 at α=0.3 when each model keeps its own ρ (Table 16 in Appendix C; DINOv2-L ×1.17). Across the interquartile range of ρ over the 12 models (0.038 to 0.341), the energy law moves AS by r_E=2.4–5.4× on these three models. The planted strength that would move AS as much, α^*, is therefore above 0.3, the largest strength we tested, on all three models. So AS does respond to null content at fixed energy. At ρ_t=0.2 and at each model's own ρ, though, content moves it less than the observed spread of energy does. This does not hold at the lowest energy (post hoc, from the same sweep). At ρ_t=0.01, α=0.3 moves AS by ×2.70 (ResNet50), ×3.47 (ResNeXt101) and ×5.26 (DINOv2-L), above r_E (2.39, 5.37 and 4.92) on 2 of the three models.

Leak-free scores and the content excess. On all 13 models, replacing each model's null content with its own leak-free surrogate, at its own energy and covariance, gives $\mathrm{AS}_{\mathrm{surr}}$ with τ=0.69 [0.39, 0.92] against SING (0.64 on S11) and Spearman 0.96 with $\mathrm{AS}_{\mathrm{SING}}$. In round 4 we pre-registered both halves of this decomposition and five more surrogates (Table 15 in Appendix C).

- *Content-free half.* AS with surrogate null content tracks SING ($\tau \geq 0.6$) for three of the five content-free surrogates. Its value depends on the surrogate: point τ on all 13 models runs from 0.49 (isotropic and mean-only) to 0.72 (zero-mean Gaussian), and $\mathrm{AS}_{\mathrm{surr}}$ with the same Gaussian tracks CLIP-space null energy κ (τ=0.92). Real null coordinates permuted across images ($\mathrm{AS}_{\mathrm{perm}}$) give 0.69 [0.39, 0.92] on all 13 and 0.64 on S11.
- *Content half.* The excess $X = \mathrm{AS}_{\mathrm{SING}}/\mathrm{AS}_{\mathrm{surr}}$ has τ=0.54 [0.14, 0.83] on all 13 and 0.60 [0.24, 0.91] on S11, and $\mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{perm}}$ has 0.62 [0.15, 0.92] and 0.71 [0.28, 1.00]. Given ρ, the partial Spearman of X with the order is 0.79 [0.02, 0.97] on S11, against 0.52 for $\mathrm{AS}_{\mathrm{perm}}$.

Both halves track the order, the outcome the pre-registration calls "cannot apportion". The data cannot apportion SING's order between null energy and image-specific null content.

4.3 Class-mean permutation

Design. Write each image's null coordinates as $z_i = \mu_{y_i} + r_i$, with μ_c the class mean over its 10 images and r_i the within-class residual. We draw a derangement π of the 1000 classes (5 draws) and set $z'_i = \mu_{\pi(y_i)} + r_i$. The logits are unchanged, and because every class has 10 images, the total null energy and the uncentred null second-moment matrix are preserved exactly (largest relative error 2×10^{-16}). Only the alignment between an image's class and its class-mean null content is broken. $\mathrm{AS}_{\mathrm{cperm}}$ is the mean AS over draws, $\Delta_{\mathrm{align}} = \mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{cperm}}$ and the share is $\Delta_{\mathrm{align}}/\mathrm{AS}_{\mathrm{SING}}$.

Result. With the released translators, $\mathrm{AS}_{\mathrm{cperm}}$ still tracks SING (τ=0.69 [0.34, 0.94] on all 13, 0.64 on E12, 0.64 on S11), and the median share is 0.10 (E12) and 0.14 (S11). With our ridge translators the pattern is the same ($\mathrm{AS}_{\mathrm{cperm}}$ 0.70 and 0.67; median share −0.04). This is the pre-registered outcome ENERGY-CARRIES under both translator families. Since both surrogates preserve null energy, this outcome was largely foreseeable once we knew that $\mathrm{AS}_{\mathrm{perm}}$ tracks the order. Δ_{align} itself tracks the order with the released translators (0.64, 0.64) but not with ours (0.09, 0.13).

Reading. We take this as evidence leaning toward energy, not as a demonstration, for two reasons. First, given ρ, the class-aligned part keeps the stronger association with the order: with the released translators the partial Spearman is 0.22 (all 13), 0.22 (E12) and 0.35 (S11) for $\mathrm{AS}_{\mathrm{cperm}}$, against 0.76, 0.81 and 0.88 for Δ_{align}. Second, Δ_{align} fails the calibration of §4.5: at fixed class content it falls as energy rises (τ=−1.00 in world C on all three bases). Our pre-registration allows no stronger claim from this test than its raw numbers.

4.4 Retrained translators

SING's translators were fit by its authors, and AS depends on them. We fit our own ridge translators (Hastie et al., 2009, §3.4) $f \mapsto$ CLIP image embedding on half of the 10k images (5000; penalty chosen on a held-out split of that half) and evaluate on the other half. Test fidelity (mean cosine) is 0.79–0.84, against 0.81–0.87 for the released translators on the same images. SING does not list the images its translators were fit on, so on our subset their fidelity may be partly in-sample; ours is measured out of sample. With retrained translators, $\mathrm{AS}_{\mathrm{SING}}$ still agrees with SING (τ=0.85 [0.61, 1.00]; 0.82 on S11; τ=0.72 against released-translator AS). The AS order is thus not specific to the released translators.

The equal-energy result is. With retrained translators, A1 gives τ=0.61 [0.15, 0.94] on E12 and 0.75 [0.32, 1.00] on S11. Read against the round-3 rule (a kill of the energy reading if τ(A1) $\geq$ 0.6 on both E12 and S11), this is at the kill level. The rule was registered for the released translators, so we apply it as a reading, not as a registered outcome. The paired CI of $\tau(\mathrm{AS}_{\mathrm{SING}}) - \tau(\mathrm{A1})$ covers 0 ([−0.20, 0.65] on E12, [−0.32, 0.49] on S11). Leaving out any two models keeps it at 0.42 or more on E12 and 0.61 or more on S11.

Released and retrained A1 order the models almost independently (τ=0.06 on E12). Retrained A1 spans a narrow range (0.048–0.091°, against 0.049–0.302° released) and follows ρ (τ=0.58 and 0.67). Given ρ, its partial Spearman with the order is 0.15 (E12) and 0.41 (S11). With these translators, matched CLIP-space energy (A4) gives 0.52 $[-0.10, 0.97]$ and 0.49, between our bars, and κ itself tracks the order (0.76, 0.75). The pre-registered penalty grid put every model at its upper edge, so we extended the grid post hoc (10^3–10^6). Over the combined grid the chosen penalties are interior: 10^4 for VGG16, VGG19 and DenseNet, and 10^3, the value already used, for the other ten models. With these penalties A1 gives 0.39 $[-0.21, 0.86]$ on E12 and 0.53 on S11. The drop from 0.61 comes from re-fitting just the two VGG models, so this number is fragile. Forcing heavier regularisation on every model (10^6) raises it to about 0.79. Whether AS per unit energy tracks SING's order therefore depends on how the translator is fit. It does not with the released translators, and it does with ours. We do not know which translator better reflects null semantics, so neither result decides between the two readings.

4.5 Calibration and translator-free scores

Semi-synthetic worlds. To check whether any estimator can tell null content from null energy, we built three worlds on each of three bases (ResNet50, ResNeXt101 and DINOv2-L, each with its released translator) by rescaling the class-mean part μ_y and the residual r of the null coordinates, with the 12 E12 values of ρ as targets. Before rescaling, we permute the residual across images. An earlier version used each image's own residual. Within a class those residuals sum to zero, so class means estimated on disjoint subsets of the class's images come out anti-correlated. The correction changes 50 of the 78 entries of Table 17 (earlier values in Table 24), but not the verdict. We measure content by L, the class-mean null energy as a share of total feature energy. In world C, L is fixed while ρ varies; in world M the residual is fixed, so L rises with ρ; in world D, ρ is fixed and L varies. An estimator is *content-diagnostic* if, on all three bases, $|\tau(\cdot, \rho)| \leq 0.2$ in world C, $\tau(\cdot, L) \geq 0.8$ in world M and $\tau(\cdot, L) \geq 0.6$ in world D. Across the eight estimators we tested (ρ, AS, A1, $\mathrm{AS}_{\mathrm{gauss}}$, X, Δ_{align}, $\mathrm{NCM}_{\mathrm{null}}$ and the selectivity R), none meets all three criteria (Table 17 in Appendix C). The bar is strict. In world C, where L is fixed and ρ grows (the absolute class-mean null energy rises $\times$1.81, since the row-space energy is held fixed), the class-mean share of null energy falls from 50% to 0.6%, and a score that responds to that share fails world C by construction. $\mathrm{NCM}_{\mathrm{null}}$ tracks L in world D (0.99–1.00) and rises with it in world M (0.44–0.88, meeting the criterion on two bases), and Δ_{align} and X track it in world D (0.91–1.00 for Δ_{align}); all three fail world C (X meets the world-C criterion on one base only). AS follows ρ at fixed content (world C, $\tau \geq 0.97$ on all bases). The energy exponents in worlds C and M run from 0.31 to 0.59, against $\hat{a}$=0.431 from the energy law and 0.53 $[0.24, 0.82]$ across models.

Further analyses (Appendix B). Three translator-free or energy-free scores (null decodability $\mathrm{NCM}_{\mathrm{null}}$, null zero-shot accuracy NZS and a selectivity ratio R) do not track SING's order (τ=0.45, 0.30 and 0.15 on E12), and none is both energy-free and fully calibrated (Appendix B.2). A pre-registered round-5 score of how the null class means are arranged relative to the class texts, $\mathrm{SO}_{\mathrm{null}}$, is exactly invariant to rescaling the null part. It passes its invariance, derangement and planted-code gates, but its world-C energy gate was testable on only one base, and its agreement with SING's order is inconclusive (τ=0.33 and 0.45; Appendix B.3). Top-variance null PCs are not clearly more semantic per unit energy than random null subspaces; the label flips with the energy exponent (Appendix B.1). A geometric decomposition shows how AS grows with energy without regard to the class text: the out-of-plane part of the removed translated component changes the angle at second order whatever its direction. The pre-registered checks of this account pass on two held-out models but are not a strong test (Appendix B.4). SING's IS also rises with null energy and tracks SING's order (τ=0.74 on all 13).

4.6 What the interventions show

Energy and SING's score cannot be separated by their agreement with SING's order (§3). AS has a content-free energy path. It grows with surrogate null content that has no image-specific content, and the out-of-plane mechanism explains how. Content-free surrogates at each model's energy reproduce the order, and so does the content excess. The data cannot apportion the order between the two. AS per unit energy loses the agreement with the released translators and keeps it with ours. Shuffling class means keeps the agreement

(τ=0.64 on E12), which leans toward energy, but given ρ the class-aligned part has the stronger association (0.81 against 0.22), and our pre-registration allows no stronger claim than these raw numbers. No estimator we tested meets all three calibration criteria, so this cannot settle the question. AS responds to planted content at fixed energy, so these results do not show that AS carries no leakage signal.

5 Do other null-space scores share the energy confound?

If null energy drives SING's order, it might also drive other published scores that read the null or residual part of a feature. In round 6 we pre-registered two new energy statistics and applied them to ViM (Wang et al., 2022), the insignificant-subspace score $S_{\rightarrow}$ of ActSub (Zöngür et al., 2025), the membership score S_{mem} of CORE (Yang et al., 2026) and the conditional $\mathcal{V}$-information that the null part adds to the logits (Hewitt et al., 2021), with a NuSA-style null share (Cook et al., 2020) as an uncounted reference. These are not the tests that found SING's confound in §3–4. `PREREG_R6.md` was pushed together with its code and synthetic unit tests before any real-model number was computed. E1 rescales every model's null part to ρ=0.005, and τ_{EQ} is Kendall τ between a score's cross-model values after and before. E2 sweeps each model through the 12 E12 values of ρ, and R_B is the range over the sweep divided by the range across the 13 models. The rule calls a score ENERGY-DRIVEN if $\tau_{\text{EQ}}<0.3$ on E12 and S11 and the median $R_B \geq 0.5$. Only Πf changes, so the logits do not, and every fitted quantity is refitted. We have no training features, so each score is fitted on 5 validation images per class. The four scores are therefore surrogates of the published ones: CORE's class means rest on fewer images than it needs (about 13 per class), ActSub's reference set is 10% of this fit half rather than of the training set, and $\mathcal{V}$-information uses a restricted probe family (Appendix G).

Positive controls (post hoc). We ran scores that the rule should flag through the identical code (Table 3; `r6/POSTHOC_R6.md`). It flags SING's IS with both translators (τ_{EQ}=0.12 and 0.24, median R_B 0.57 released and 0.79 retrained). It misses the other three controls. The NuSA share moves fully with the sweep (R_B=0.89) but keeps part of its order after E1 (τ_{EQ}=0.27 and 0.31), because the per-image null share keeps a model-specific spread when the mean energy is equalised. ρ itself has no order left after E1, so τ_{EQ} is undefined and the rule returns mixed. SING's AS with the released translators gives τ_{EQ}=0.39 and 0.45 and median R_B=0.26 (mixed), and with the retrained translators 0.85, 0.85 and 0.67 (mixed). A rule that misses SING's AS and a pure energy score cannot show that other scores are free of the confound. The pre-registered count (k=0 of 4 scores energy-driven) is therefore uninformative, and we do not use it to decide whether the confound is specific to SING. Yet equalising energy leaves released AS with only 0.11 of its cross-model range (below), so the miss comes from the thresholds, not from weak energy dependence. The τ part of the rule also says little by itself: if the order after E1 were unrelated to the order before, $\tau_{\text{EQ}}<0.3$ would hold with probability 0.90 for 12 models.

What the statistics show (descriptive). ViM, ActSub and CORE keep their order under E1 (τ_{EQ} 0.78–0.89), but all three are scale-normalised by construction. ViM refits α, the ratio of mean maximum logit to mean residual norm, on every fit, so rescaling the null part rescales α inversely. With α frozen at its real-data value, E1 moves ResNet50's ViM score from 0.876 to 2.5×10^{-4} (refitting multiplies α by 0.8 to 8.5 across models). ActSub's cosine neighbours and CORE's cosine scores do not depend on scale. As a magnitude measure (post hoc), the share of the cross-model range of the E12 models left after E1 is 5.4×10^{-3} for the NuSA share, 0.11 for SING's AS (released), 0.29 for $\mathcal{V}$-information, 0.78 for ActSub, 0.90 for ViM (1.93 with α frozen) and 0.92 for CORE. In the semi-synthetic worlds of §4.5, ActSub, CORE and $\mathcal{V}$-information fall as ρ grows at fixed content (world C, τ_ρ from -0.73 to -1.00 on all three bases), ViM rises with it ($+1.00$), and ViM moves against content in worlds M and D (τ_L down to -1.00; Table 27). $\mathcal{V}$-information depends most on energy of the four, but 86% of its cross-model range comes from DenseNet121, four of its 13 gains are zero to four decimals or negative, and its penalty sat at the top of the grid in 15 of 26 fits (Appendix G). None of the four tracks SING's order (Table 26). We do not read the content tests. Their within-class permutation moves every score by more than 1 SD on several models, and a synthetic check shows that this permutation can remove real image-level conditional information while keeping each class's null distribution (a gain of 1.18 nats falls to -9.7×10^{-3}). The flag therefore cannot separate a real loss of within-class or image-level information from estimator noise.

Score	τ_{EQ} E12	τ_{EQ} S11	med. R_B	range left after E1	rule output
Positive controls (energy scores or SING's own score; post hoc except NuSA)					
NuSA share	0.27	0.31	0.89	5.4×10^{-3}	mixed
ρ itself	–	–	0.99	0	mixed
SING AS, released	0.39	0.45	0.26	0.11	mixed
SING AS, retrained	0.85	0.85	0.67	0.15	mixed
SING IS, released	0.12	0.24	0.57	0.11	energy-driven
SING IS, retrained	0.12	0.24	0.79	0.05	energy-driven
Audited scores (pre-registered; frozen-α ViM post hoc)					
ViM	0.88	0.89	0.06	0.90	not energy-driven
ViM, α frozen†	-0.52	-0.49	1.33	1.93	energy-driven
ActSub $S_{\rightarrow}$	0.79	0.78	0.14	0.78	mixed
CORE S_{mem}	0.85	0.82	0.11	0.92	mixed
$\mathcal{V}$-information	0.20	0.33	0.19	0.29	mixed

Table 3: Round-6 energy statistics with post hoc positive controls (§5). τ_{EQ}: Kendall τ between a score after equalising null energy (E1) and before, on the estimable models. Median R_B: range over the energy sweep E2 relative to the range across models. Range left after E1: range(E1)/range(real) over the estimable E12 models (post hoc). Rule output: ENERGY-DRIVEN needs $\tau_{\text{EQ}}<0.3$ on both sets and median $R_B\geq0.5$; NOT ENERGY-DRIVEN needs $\tau_{\text{EQ}}\geq0.6$ and $R_B\leq0.1$; anything else is mixed. For ρ, τ_{EQ} is undefined because all E1 values are equal. †ViM with α frozen at its real-data value of each fit half. Pre-registered content counts and labels: Table 25 (Appendix G).

6 The DinoViT checkpoint

SING reports its "DinoViT" at 84.0% top-1 and itself describes the model as pretrained beyond ImageNet. Its released configuration loads a timm AugReg ViT-L/16. We evaluated five public ViT-L/16 checkpoints on the full 50k validation set: torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear (Singh et al., 2022) (85.14%), SWAG end-to-end (88.07%) and `google/vit-large-patch16-224` (81.98%) (Fig. 7, Appendix D). None falls within 84.0 ± 0.3. SING's evaluation protocol is not specified, so the 84.0% versus 85.85% gap is unexplained.

With SING's `dinovit1` translator on 1000 images, the timm AugReg features give mean cosine 0.848 to the CLIP image embeddings. Their shuffled-pair floor is 0.575, centred cosine is 0.674 and retrieval top-1 is 78.8%. Every other candidate, re-extracted with its native transform, is at its own shuffled floor (largest gap 0.002) with retrieval top-1 at most 0.2%. SING's translator was thus fit on this supervised timm checkpoint, while the other candidates are at chance. DINO never released a ViT-L with a 1000-way head (Caron et al., 2021). SING's narrative about self-supervision should therefore not rest on this model.

True DINO backbones. The two DINO backbones released with a linear ImageNet head, ViT-B/16 and ViT-S/16 (Caron et al., 2021), have no SING translator, so we use ridge translators fit as in §4.4. By raw AS they rank 8 and 9 of 15, which by the pre-registered raw-AS question is "SING-inconsistent". Raw ridge AS also tracks null energy, which is above the median for both. Against the energy law fitted on E12, ViT-B/16 lies inside the 95% prediction interval and ViT-S/16 just below it. That leans toward SING's account for ViT-S/16 and leaves the evidence mixed (Appendix B.5, Table 9). Both DINO heads are linear probes on frozen features, so the comparison is indicative only.

7 Size-corrected scores and SING's claims

Dividing AS by the AS of an energy-matched random removal, AS_{fair}, over three families of controls gives point τ against SING of -0.18, -0.41 and -0.13 on all 13 models, all below the pre-registered 0.3 bar. Over eight normalisers, the pre-registered reversal criterion (CI entirely below 0 under at least 5 of 8) fails (0/8 on all 13; on S11 only family B, which is not logit-preserving, meets it). Family-A AS_{fair} over-corrects (Spearman with ρ -0.79 on S11). Size-corrected rankings therefore *differ* from SING's, but the direction depends on

the normaliser, and we make no reversal claim. For SING's testable class-level claims under $\mathrm{AS}_{\mathrm{SING}}$, 7 hold, 3 reverse (one borderline) and 1 is contradicted; these differences may reflect protocol differences and are not caused by size correction (Appendix B.6, Table 10).

8 Discussion

What we can and cannot conclude. SING's cross-model ranking is indistinguishable from the ranking by null energy. AS responds to null energy through a path that carries no image-specific content, and we can describe that path geometrically. The data cannot separate null energy from real null content as the source of SING's order: content-free surrogates and the content excess both reproduce it, AS per unit energy tracks the order with one translator family and not the other, the class-mean permutation leans toward energy without settling it (its raw numbers are all our pre-registration supports), no estimator we tried meets all three calibration criteria, and a pre-registered score of null class organisation is inconclusive against the order. We do not conclude that SING's ranking is wrong, or that AS carries no semantic signal; AS responds to planted concepts at fixed energy. The claim we support is narrower: a cross-model AS comparison cannot be read as a comparison of null semantics, because energy and content are not separated by any control we have. Whether the confound extends to other null-space scores is open. The round-6 energy rule flags SING's IS but not its AS or a pure energy score (§5), so its verdicts on ViM, ActSub, CORE and $\mathcal{V}$-information cannot settle the question. Three of those scores keep their order when null energy is equalised largely because they normalise scale by construction. Those scores were built for OOD detection rather than for comparing null semantics across models, and none of them tracks SING's order (ActSub and CORE are inconclusive).

Why does ρ vary? Neural collapse (Papyan et al., 2020) offers a candidate mechanism. Late in training, features concentrate near class means aligned with the classifier rows, so the energy outside the row space shrinks. Models trained further into that regime, or with heads fit differently, would have smaller ρ and hence smaller AS. We have not tested this.

Relation to other work. Similar negative controls appear in other areas. Saliency maps that survive model and data randomisation fail as explanations (Adebayo et al., 2018), attention weights need adversarial and random baselines before they can be read as explanation (Wiegreffe & Pinter, 2019), and apparent emergent abilities can be an artefact of the metric rather than the model (Schaeffer et al., 2023). Our surrogates are the null-space analogue of the surrogate-data test (Theiler et al., 1992): replace the content while keeping the second moments, and see whether the statistic changes. Cui et al. (2022) regress a confound out of representation-similarity scores such as RSA and CKA. Gröger et al. (2026) show that representational similarity metrics are confounded by model scale, calibrate them with permutation nulls, and find that global convergence largely disappears while local neighbourhood similarity survives, a mixed picture of the same shape as ours. Both argue, as we do, that a cross-model score needs a baseline for a nuisance variable before it is read as a property of the models. Makelov et al. (2024) show that activation patching along a subspace can change behaviour through a dormant pathway that the model does not normally use; their illusion and our energy path both arise when a component the model ignores still moves a readout. In OOD detection, Mahalanobis++ (Müller & Hein, 2025) finds that feature norms confound Mahalanobis scores across 44 models and that ℓ_2 normalisation mitigates it, a feature-energy confound in a different kind of score, and Park et al. (2023) analyse how the feature norm acts as a hidden OOD signal.

Relation to other null-space work. ActSub (Zöngür et al., 2025) finds that the activation components a classifier largely ignores help to detect distribution shift, and Jacobsen et al. (2019) show that class information can hide in directions the logits do not use. Both are consistent with null content being informative. Cross-model scores of that content, however, need an energy-only baseline. Linear erasure methods (Ravfogel et al., 2020; Belrose et al., 2023) remove a concept so that no linear probe can recover it, a stricter notion than projecting out $\ker W$. A leakage score built on them would ask whether class information is linearly recoverable from the null component rather than how far the translation moves. Linear maps between representations go back to model stitching (Lenc & Vedaldi, 2015; Bansal et al., 2021). Linear translators into CLIP (Moayeri et al., 2023) differ in fidelity across models, and §4.4 shows that the translator affects

interventional conclusions. XCC is a cross-fitted form of centred kernel alignment between class Gram matrices (Cortes et al., 2012; Kornblith et al., 2019). $\mathrm{SO}_{\mathrm{null}}$ differs from class-separation scores such as the R^2 of Kornblith et al. (2021), Fisher-type between/within ratios and the within-class variability of neural collapse (Papyan et al., 2020): those measure how much class structure a representation holds, whereas $\mathrm{SO}_{\mathrm{null}}$ measures how that structure is arranged relative to the class texts, so class structure unrelated to the texts lowers it.

Recommendations.

1. Report d_{null}, ρ and the null tolerance next to any null-space score, and check rank stability across tolerances.
2. Compare every cross-model AS-type score (a score that measures how far a readout moves when the null component is removed) with the ranking by ρ alone, as a control in the sense of Hewitt & Liang (2019), and with content-free surrogates of matched energy. Report the surrogate used, since the result depends on it.
3. Check conclusions with a second, independently fit translator, and calibrate any content estimator on worlds with known content before using it across models.
4. Name the exact checkpoint and its measured accuracy, and check translator fidelity against a shuffled floor.
5. Flag translator conditioning when full-null and partial-null AS disagree (BiT).
6. Before reading a null-space score across models, rescale the null part to equal energy and sweep it through the observed range (E1 and E2 of §5). Check the statistics on positive controls (ρ, an AS-type score, and the score with any built-in normalisation frozen), and report the share of the cross-model range left after equalisation. Both tests are cheap and leave the logits unchanged.
7. Validate flatness or invariance criteria on a synthetic oracle before using them to accept or reject scores (Appendix E).

Limitations. There are only 13 models, so every cross-model CI is wide. SING's order is digitised from a figure. Our retrained translators are ridge maps chosen by us. SING's training recipe may differ, and the equal-energy result depends on this choice. The energy-scan part of the leak-free test covers three models with d_{null}=1048, and the planted strengths stop at α=0.3, below the energy-equivalent strength at each model's own ρ (at ρ_t=0.01, post hoc, content moves AS more than energy does on 2 of the three models). The calibration worlds are semi-synthetic, built on three bases by rescaling class-mean and residual null content; they test whether an estimator can separate the two, not whether real models differ in content. The organisation score's thresholds were set after scratch runs on five models (SO) and three (XCC), two of its bars sit at the scratch values, and its world-C gate was testable on one base only, over ρ=0.005–0.013. The geometric account is consistent with two held-out models and with synthetic data that builds in its gain law; on the 13 models it is exploratory. The true DINO comparison uses our ridge translators, since SING released none for these models. The energy law does not cleanly separate energy from dimension, and its pre-registered flatness part fails. The class-claim CIs reflect image resampling only (class resampling for C6), several claims rely on proxies, and C13 is untestable. The round-6 tests fit every score on 5 validation images per class instead of the training set, so the four scores are surrogates of the published ones. They use a restricted probe family for $\mathcal{V}$-information and cannot run CANARY's per-class neighbour rule (Shin & Kim, 2026). Their energy rule misses three of its five post hoc positive controls (it flags only SING's IS), and their content tests cannot separate a loss of image-level information from estimator noise. We did not evaluate information-theoretic leakage measures for concept models (Makonnen et al., 2025), which target a different notion of leakage.

9 Conclusion

Using SING's code and released translators, we reproduce its 13-model ranking, and the null energy fraction ρ reproduces it equally well. A content-free path connects null energy to AS, and image-independent null

content with each model's own energy reproduces the order. The content excess reproduces it too, and AS per unit energy tracks the order with retrained translators but not with the released ones, so the data cannot separate null energy from real null content as the source of the order. SING's "DinoViT" translator matches a supervised timm checkpoint. Under our ridge translators, true DINO backbones fall mid-table by raw AS, and the evidence is mixed once energy is accounted for. Size-corrected scores change the order without reversing it with confidence. Our pre-registered energy rule for other null-space scores flags SING's IS but not its AS or a pure energy score, so it cannot say whether the confound extends to them. Cross-model AS-type null-space comparisons should report ρ, an energy-only baseline and content-free surrogates of matched energy next to the score.

Reproducibility. The supplementary material contains the code, pre-registrations, aggregated results and a one-command `make reproduce` that regenerates every table, figure and number in this paper from committed aggregates. For review, we provide the code as an anonymised archive, and a public link will be added on acceptance. ImageNet-derived features are not redistributed; checksums, the image list and the extraction scripts are included. A full rerun from features takes about 2 h on 8 CPU cores; feature extraction needs a GPU and ImageNet (Appendix H).

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, pp. 9525–9536, 2018.

Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2306.03819.

Mouïn Ben Ammar, Nacim Belkhir, Sebastian Popescu, Antoine Manzanera, and Gianni Franchi. NECO: NEural collapse based out-of-distribution detection. In *International Conference on Learning Representations (ICLR)*, 2024.

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9650–9660, 2021.

Matthew Cook, Alina Zare, and Paul Gader. Outlier detection through null space analysis of neural networks. *arXiv preprint arXiv:2007.01263*, 2020.

Corinna Cortes, Mehryar Mohri, and Afshin Rostamizadeh. Algorithms for learning kernels based on centered alignment. *Journal of Machine Learning Research*, 13(28):795–828, 2012.

Tianyu Cui, Yogesh Kumar, Pekka Marttinen, and Samuel Kaski. Deconfounded representation similarity for comparison of neural networks. In *Advances in Neural Information Processing Systems*, 2022.

MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf, and Eugene Belilovsky. Reliability of CKA as a similarity measure in deep learning. In *International Conference on Learning Representations (ICLR)*, 2023.

Bradley Efron. Better bootstrap confidence intervals. *Journal of the American Statistical Association*, 82 (397):171–185, 1987.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021.

Fabian Gröger, Shuo Wen, and Maria Brbić. Revisiting the platonic representation hypothesis: An Aristotelian view. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 36724–36766, 2026.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2nd edition, 2009.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 2733–2743, 2019.

John Hewitt, Kawin Ethayarajh, Percy Liang, and Christopher D. Manning. Conditional probing: Measuring usable information beyond a baseline. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021. arXiv:2109.09234.

Daksh Idnani, Vivek Madan, Naman Goyal, David J. Schwab, and Shanmukha Ramakrishna Vedantam. Don't forget the nullspace! Nullspace occupancy as a mechanism for out of distribution failure. In *International Conference on Learning Representations (ICLR)*, 2023. OpenReview id 39z0zPZ0AvB.

Jörn-Henrik Jacobsen, Jens Behrmann, Richard Zemel, and Matthias Bethge. Excessive invariance causes adversarial vulnerability. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1811.00401.

Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big transfer (BiT): General visual representation learning. In *European Conference on Computer Vision (ECCV)*, pp. 491–507, 2020.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 3519–3529, 2019.

Simon Kornblith, Ting Chen, Honglak Lee, and Mohammad Norouzi. Why do better loss functions lead to less transferable features? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008. doi: 10.3389/neuro.06.004.2008.

Karel Lenc and Andrea Vedaldi. Understanding image representations by measuring their equivariance and equivalence. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 991–999, 2015.

Xiang Li and Kevin M. Short. Null space properties of neural networks with applications to image steganography. *arXiv preprint arXiv:2401.10262*, 2024.

Aleksandar Makelov, Georg Lange, and Neel Nanda. Is this the subspace you are looking for? An interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024.

Mikael Makonnen, Moritz Vandenhirtz, Sonia Laguna, and Julia E. Vogt. Measuring leakage in concept-based methods: An information theoretic approach. *arXiv preprint arXiv:2504.09459*, 2025.

Mazda Moayeri, Keivan Rezaei, Maziar Sanjabi, and Soheil Feizi. Text-to-concept (and back) via cross-model alignment. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, 2023. arXiv:2305.06386.

Maximilian Müller and Matthias Hein. Mahalanobis++: Improving OOD detection via feature normalization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 45151–45184. PMLR, 2025.

Hamed Nili, Cai Wingfield, Alexander Walther, Li Su, William Marslen-Wilson, and Nikolaus Kriegeskorte. A toolbox for representational similarity analysis. *PLoS Computational Biology*, 10(4):e1003553, 2014. doi: 10.1371/journal.pcbi.1003553.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. arXiv:2304.07193.

Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020. arXiv:2008.08186.

Jaewoo Park, Jacky Chen Long Chai, Jaeho Yoon, and Andrew Beng Jin Teoh. Understanding the feature norm for out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pp. 8748–8763, 2021.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 7237–7256, 2020.

Hossein Rezaei and Mohammad Sabokrou. Quantifying overfitting: Evaluating neural network performance through analysis of null space. *arXiv preprint arXiv:2305.19424*, 2023.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems*, 2023.

Wonseok Shin and Songkuk Kim. A canary in the neural mine: Catching an image classifier's silent, confident failures. *arXiv preprint arXiv:2610.33063*, 2026.

Mannat Singh, Laura Gustafson, Aaron Adcock, Vinicius de Freitas Reis, Bugra Gedik, Raj Prateek Kosaraju, Dhruv Mahajan, Ross Girshick, Piotr Dollár, and Laurens van der Maaten. Revisiting weakly supervised pre-training of visual perception models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 804–814, 2022. arXiv:2201.08371.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.

James Theiler, Stephen Eubank, André Longtin, Bryan Galdrikian, and J. Doyne Farmer. Testing for nonlinearity in time series: The method of surrogate data. *Physica D: Nonlinear Phenomena*, 58(1–4): 77–94, 1992. doi: 10.1016/0167-2789(92)90102-S.

Hugo Touvron, Matthieu Cord, and Hervé Jégou. DeiT III: Revenge of the ViT. In *European Conference on Computer Vision (ECCV)*, pp. 516–533, 2022.

Alexander Walther, Hamed Nili, Naveed Ejaz, Arjen Alink, Nikolaus Kriegeskorte, and Jörn Diedrichsen. Reliability of dissimilarity measures for multi-voxel pattern analysis. *NeuroImage*, 137:188–200, 2016. doi: 10.1016/j.neuroimage.2015.12.012.

Haoqi Wang, Zhizhong Li, Litong Feng, and Wayne Zhang. ViM: Out-of-distribution with virtual-logit matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4921–4930, 2022.

Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 11–20, 2019. doi: 10.18653/v1/D19-1002.

Harel Yadid, Meir Yossef Levi, Roy Betser, and Guy Gilboa. Make it SING: Analyzing semantic invariants in classifiers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9911–9920, 2026. arXiv:2603.14610 (v2 supplement cited for Supp. Table 3 and Supp. Fig. 12b).

Jin Mo Yang, Hyung-Sin Kim, and Saewoong Bahk. CORE: Robust out-of-distribution detection via confidence and orthogonal residual scoring. *arXiv preprint arXiv:2603.18290*, 2026.

Barış Zöngür, Robin Hesse, and Stefan Roth. Activation subspaces for out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2508.21695.

A Pre-registration ledger

Table 4 lists every pre-registered experiment and its outcome. E1 to E9 come from the original plan, R2 from the round-2 pre-registration and R3 from the round-3 one, each written after the previous round's results. Round 4 (R4, R4B and R4C, three files) is in Table 5; the addenda R4B and R4C were written after the R4 results, and each lists the values known when it was written. Round 5 (`PREREG_R5.md`) is in Table 6, and round 6 (`PREREG_R6.md`) in Table 7.

Exp.	Pre-declared criterion (short)	Outcome
Exp 0	null size correlates with SING IS/AS	GO rests on ρ (Spearman +0.94 with the leakage order); d_{null} alone +0.46, p=0.12
Exp 1	r=AS(1048)/AS(24) ≥ 3 on $\geq 3/4$; $\tau_{d=24} \leq 0.5$	GO ($r \approx 10$ on 3/4; BiT 0.06; τ=0.29)
E1	high/low $\geq 3\times$ at fixed d; $\leq 1.5\times$ across d at fixed ρ	MIXED (first part 4/4; second fails)
E2	$\tau(\text{AS}_{\text{fair}}, \text{SING}) \leq 0.5$ or headline claim reverses	PASS as registered; weakened in round 2
E3	$\tau(\text{AS}_{\text{fair}}, \text{planted}) \geq 0.8$, $\tau(\text{AS}_{\text{SING}}, \text{planted}) \leq 0.5$	not passed (0.109 vs 0.386); KILL not met
E8	NCP probe $\tau \geq 0.8$ and flat; KILL if $\tau < 0.6$	KILL (τ=−0.331; flat 0/4)
E9	20-candidate screen, DEV $\tau \geq 0.8$ and flat on $\geq 3/4$	KILL (best M1 τ=0.976, flat 0/4)
R2	KILL-R: drop "disagrees" unless fix 2 and D-weak pass	not fired; "differs from" allowed; D-weak screened at the threshold (5/8)
R2	D-strong: τ CI < 0 under $\geq 5/8$ normalisers	FAIL (0/8 all 13; 1/8 screened)
R2	AS_{fair} removes the ρ trend	FAIL on screened set (-0.79 [$-1.00, -0.23$])
R2	tolerance stability: adjacent $\tau \geq 0.9$	FAIL (AS_{fair} 0.87/0.82; AS_{SING} 0.87); $\rho \geq 0.92$
R2	PC bands $G_{\text{top}}/G_{\text{rand}}$ vs 1 at d=24	"top PCs carry less AS per unit energy" (2/3 screened models)
R2	synthetic oracle passes the flatness rule	FAIL (ratio 3.55): rule ill-posed
R2	DinoViT translator fidelity: used checkpoint best by ≥ 0.05	PASS (0.848 vs 0.677); see D
R3-A	equal-energy A1: ENERGY-ACCOUNTS / PARTIAL / KILL-R3	ENERGY-ACCOUNTS ($\tau = -0.03$ E12, 0.09 S11); no KILL
R3-A	A4 (matched CLIP energy) $\tau < 0.3$ on both sets	yes (-0.15, -0.20); $\kappa - \rho$ CI covers 0
R3-B	N4: AS rises with leak-free null energy, all 3 models	PASS (Spearman 1; slopes 0.38–0.69)
R3-B	B-rank: $\tau(\text{AS}_{\text{surr}}) \geq 0.6$ on all 13 and S11	yes (0.69, 0.64); no claim from X
R3-C	PC bands vs energy law $\rho^{\hat{a}}$	"top PCs carry less AS than energy predicts"; not robust in $\hat{a}$
R3-D	N3: non-matching candidates at chance	PASS ($\lvert\cos - \text{shuffled}\rvert \leq 0.002$)
R3-E	N7: no claim CI excludes its point	FAIL (1 flag: AS_{fair} C6 SD); N6 dedup done
R3-F	N1: paired $\tau(\rho) - \tau(\text{AS}_{\text{SING}})$ CI covers 0	yes $[-0.36, 0.29]$: "indistinguishable"
R3-G	retrained translators: order survives if $\tau \geq 0.6$ on all 13 and S11	survives (0.85, 0.82); A1 with retrained translators 0.61, 0.75 (descriptive)

Table 4: All pre-registered experiments of rounds 1–3 and their outcomes. The pre-registration files are in the repository.

Exp.	Pre-declared criterion (short)	Outcome
R4-S	content-free AS_{perm} and content excess $AS_{SING} - AS_{perm}$ each track SING ($\tau \geq 0.6$ on all 13 and S11, CI > 0); KILL-R4-S if $\tau(AS_{perm}) < 0.3$	both track: cannot apportion (0.69/0.64 and 0.62/0.71); surrogate τ spans 0.49–0.72; no kill
R4-J	KILL-R3 rule applied to retrained A1 (reading only); influence of DinoViT/EVA02	retrained: KILL-R3 level (0.61, 0.75); released S11 without the pair 0.61
R4-J	A4 under ridge translators < 0.3 on both sets	no (0.52, 0.49): holds for released translators only
R4-DN	KILL-DN (feature validity); Q-DN1 (DINO among lowest 4 AS); Q-DN2 (inside energy-law PI)	no kill; Q-DN1 SING-inconsistent (ranks 8, 9 of 15); Q-DN2 mixed
R4-P	energy-equivalent planted strength α^*	$\alpha^* > 0.3$ on all 3 models (outside the tested range)
R4-C6	C6-SD point inside both CIs (class resampling); no flags	PASS
R4-N10	fast holdout < 30 min and within ± 0.03	FAIL (36.3 min; ΔAS_{fair} nl 0.057); not re-timed, see Appendix H
R4B-1	class-mean permutation: CONTENT-CARRIES / ENERGY-CARRIES / UNAPPORTIONED; KILL-R4B-1 on CONTENT-CARRIES	ENERGY-CARRIES under both families (released 0.64/0.64; ridge 0.70/0.67); no kill
R4B-2	semi-synthetic worlds C/M/D: content-diagnostic estimator	none of 8 estimators diagnostic on all 3 bases
R4B-5	NCM_{null}: NCM-CONTENT (≥ 0.6, partial CI $\not\ni 0$) / NCM-NULL (< 0.3)	intermediate (0.45, 0.42); no kill
R4B-3	selectivity R: R-TRACKS (≥ 0.6) / R-NULL (< 0.3); within-model CI $\not\ni 1$ on $\geq 9/12$	R-NULL (0.15, 0.05); within-model 12/12 (4 above, 8 below 1)
R4B-6	DINO inside the E12 PI of Δ_{align} and NCM_{null}; both in the lowest 4	inside both PIs for both; 'least leaking' label not met (it needs both models in the lowest 4); ViT-S/16 alone is in the lowest 4 on both (ranks 3 of 15 and 2 of 14)
R4C-G2	geometry on true DINO: (i) Spearman ≥ 0.9, (ii) oop CI > 0, (iii) closed form $\leq 25\%$; (iv) $\lvert a_{fit} - a_{pred} \rvert \leq 0.1$	PASS on both; exponent predicted; no kill
R4C-G3	synthetic: a_{fit} within ± 0.05 of $(1+s)/2$	PASS (max error 0.027)
R4C-B3	NZS gate (rescaling $\leq 5\%$, planted $\tau \geq 0.6$); energy-free $\lvert \tau(\cdot, \rho) \rvert < 0.3$; SING test	gate pass; not energy-free (0.45, 0.53); SING test inconclusive
R4C-B4	IS energy proxy: $\lvert \tau(\text{IS}, 1-\rho) \rvert \geq 0.6$ on all three sets	no (-0.62, -0.55, -0.49)

Table 5: Round-4 pre-registered tests and outcomes (E12 / S11 unless stated). R4B and R4C rows come from addenda written after the R4 results. No kill fired.

Exp.	Pre-declared criterion (short)	Outcome
R5-cal	world residuals permuted across images (seed 99000); report every changed value	50 of 78 values change; content-diagnostic estimators: none before, none after
R5-G1	exact invariance to $\beta \in \{0.1, 10\}$ and orthogonal maps	SO pass, XCC pass
R5-G2	class-mean derangement: fail on ≤ 2 models	fails: SO 0, XCC 0 (max |mean| 0.002, 0.001)
R5-G3	world S: semantic ≥ 2 SD, arbitrary ≤ 1 SD on $\geq 11/13$	SO 13/13, XCC 13/13
R5-G4	corrected world C: max / min ≤ 1.5; kill if it fails on ≥ 2 bases	SO testable on one base (pass): not testable; XCC testable on two, passes on two
R5-G5	estimable on ≥ 9 of E12 and ≥ 8 of S11	SO 12/11, XCC 12/11; no kill
R5-SO	ORGANISATION-TRACKS ($\tau \geq 0.6$ on E12 and S11) / DOES-NOT-TRACK (< 0.3 on both) / INCONCLUSIVE	INCONCLUSIVE (0.33 [-0.29,0.79], 0.45 [-0.21,0.88])
R5-XCC	same bars, secondary	DOES-NOT-TRACK (-0.12, -0.05)
R5-inh	INHERITED if F1-partial ≤ 0.05 on $\geq 75\%$; plant check ≥ 0.1	NOT INHERITED (7/13, 54%); plant check pass

Table 6: Round-5 pre-registered tests and outcomes (E12 / S11 unless stated). Thresholds were set after scratch runs on five models (SO-type) and three (XCC-type), with plants, worlds and derangements, and a ResNet50 smoke test of the final code. No kill fired.

Test	Criterion	Outcome
G5 estimability	$\mathrm{SD}_{\mathrm{boot}} \leq 0.1\times$ range; verdict needs $\geq$9 E12 and $\geq$8 S11	ViM: 12/11; ActSub $S_{\rightarrow}$: 12/11; CORE S_{mem}: 12/11; $\mathcal{V}$-info (class): 11/10
ViM, energy	ENERGY-DRIVEN if $\tau_{\mathrm{EQ}} < 0.3$ on both sets and median $R_B \geq 0.5$; NOT if ≥ 0.6 and ≤ 0.1	NOT ENERGY-DRIVEN (τ_{EQ} 0.88 / 0.89, R_B 0.06)
ViM, content	RESPONSIVE if $\lvert C_1\rvert \geq 2$ and $\lvert\Delta_{\mathrm{arb}}\rvert \geq 2$ on $\geq$10 of 13; BLIND if both ≤ 1 on $\geq$10	partial (9 / 5 of 13)
ActSub $S_{\rightarrow}$, energy	ENERGY-DRIVEN if $\tau_{\mathrm{EQ}} < 0.3$ on both sets and median $R_B \geq 0.5$; NOT if ≥ 0.6 and ≤ 0.1	MIXED (τ_{EQ} 0.79 / 0.78, R_B 0.14)
ActSub $S_{\rightarrow}$, content	RESPONSIVE if $\lvert C_1\rvert \geq 2$ and $\lvert\Delta_{\mathrm{arb}}\rvert \geq 2$ on $\geq$10 of 13; BLIND if both ≤ 1 on $\geq$10	partial (11 / 8 of 13)
CORE S_{mem}, energy	ENERGY-DRIVEN if $\tau_{\mathrm{EQ}} < 0.3$ on both sets and median $R_B \geq 0.5$; NOT if ≥ 0.6 and ≤ 0.1	MIXED (τ_{EQ} 0.85 / 0.82, R_B 0.11)
CORE S_{mem}, content	RESPONSIVE if $\lvert C_1\rvert \geq 2$ and $\lvert\Delta_{\mathrm{arb}}\rvert \geq 2$ on $\geq$10 of 13; BLIND if both ≤ 1 on $\geq$10	partial (11 / 8 of 13)
$\mathcal{V}$-info (class), energy	ENERGY-DRIVEN if $\tau_{\mathrm{EQ}} < 0.3$ on both sets and median $R_B \geq 0.5$; NOT if ≥ 0.6 and ≤ 0.1	MIXED (τ_{EQ} 0.20 / 0.33, R_B 0.19)
$\mathcal{V}$-info (class), content	RESPONSIVE if $\lvert C_1\rvert \geq 2$ and $\lvert\Delta_{\mathrm{arb}}\rvert \geq 2$ on $\geq$10 of 13; BLIND if both ≤ 1 on $\geq$10	content-responsive (11 / 12 of 13)
Counting rule	GENERAL if $k \geq 3$ (n=4) or $k \geq 2$ (n=3); DESIGN-DEPENDENT if n=4, k=2; SING-SPECIFIC if $k \leq 1$	n=4, k=0: SING-SPECIFIC
Content measurable	$\geq$2 counted scores NOT ENERGY-DRIVEN and RESPONSIVE	no (0)

Table 7: Round-6 pre-registered verdicts (`PREREG_R6.md`, pushed with its code and synthetic unit tests before any real-model number; the content labels are registered outcomes, not readings, see Appendix G). τ_{EQ} on E12 / S11; content counts are $\lvert C_1\rvert \geq 2$ / $\lvert\Delta_{\mathrm{arb}}\rvert \geq 2$ out of 13 models.

B Details of further analyses

This appendix holds analyses summarised in the main text, unchanged.

Round-4 scores. Table 8 lists the round-4 scores against SING's order.

B.1 Are high-energy null directions more semantic?

If the high-variance null directions were the semantic ones, energy would partly *be* the signal. At matched d, we compare the top-variance null PCs with random null subspaces. In round 2 we used $G = \mathrm{AS}/\sqrt{\rho}$. The pre-registered round-2 label was "top PCs carry less AS per unit energy" on 2 of the 3 screened models at d=24: $G_{\mathrm{top}}/G_{\mathrm{rand}}$ was 0.60 for ResNeXt101 and 0.87 for DINOv2-L, with CIs below 1, against 1.16 for ResNet50. At d=100 the ratios were 1.01 (ResNet50), 0.85 (ResNeXt101), 0.83 (DINOv2-L) and 0.011 (BiT, not counted). But $\sqrt{\rho}$ is not the fitted law. In round 3 we compare against $\rho^{\hat{a}}$, $H = (\mathrm{AS}_{\mathrm{top}}/\rho_{\mathrm{top}}^{\hat{a}})/(\mathrm{AS}_{\mathrm{rand}}/\rho_{\mathrm{rand}}^{\hat{a}})$. At $\hat{a}$=0.431 and d=24, H is 1.50 (ResNet50), 0.74 (ResNeXt101) and 0.98 [0.97, 0.99] (DINOv2-L), which gives the pre-registered label "top PCs carry less AS than energy predicts" (2/3). DINOv2-L's H differs from 1 by 2%, a negligible effect even though its CI excludes 1. At the low end of the CI for a (0.388), the label flips to "more", so it is *not robust* (Table 13 in Appendix C). High-energy null directions are not clearly more semantic per unit energy. The result varies across models and with the exponent, so this test does not decide between the readings.

B.2 Translator-free and energy-free scores

- *Null decodability* $\mathrm{NCM}_{\mathrm{null}}$: leave-one-out nearest-class-mean accuracy on the normalised null coordinates, minus a shuffled-label control. It needs no translator. It ranges from 5.2 to 80.9 points and gives τ=0.45 $[-0.11, 0.90]$ on E12 and 0.42 on S11, between our bars. It follows ρ (0.48, 0.42), and its partial association given ρ is about zero (-0.05, -0.01).

Score	E12	S11
A1 and its influence (released)		
A1, all models	-0.03 $[-0.54, 0.61]$	0.09 $[-0.45, 0.78]$
A1 without DinoViT and EVA02	0.38	0.61
Retrained ridge translators		
A1	0.61 $[0.15, 0.94]$	0.75 $[0.32, 1.00]$
A4 (matched CLIP-space energy)	0.52 $[-0.10, 0.97]$	0.49 $[-0.17, 1.00]$
κ	0.76 $[0.40, 0.97]$	0.75 $[0.32, 1.00]$
Class-mean permutation (idea 1)		
$\mathrm{AS}_{\mathrm{cperm}}$, released	0.64 $[0.28, 0.90]$	0.64 $[0.22, 0.92]$
Δ_{align}, released	0.64 $[0.28, 0.90]$	0.64 $[0.24, 0.91]$
$\mathrm{AS}_{\mathrm{cperm}}$, ridge	0.70 $[0.33, 0.94]$	0.67 $[0.24, 1.00]$
Δ_{align}, ridge	0.09 $[-0.57, 0.61]$	0.13 $[-0.58, 0.63]$
Translator-free and energy-free scores		
$\mathrm{NCM}_{\mathrm{null}}$ (idea 5)	0.45 $[-0.11, 0.90]$	0.42 $[-0.22, 0.92]$
null zero-shot NZS (B3)	0.30 $[-0.31, 0.89]$	0.38 $[-0.28, 0.96]$
selectivity R, k=20 (idea 3)	0.15 $[-0.44, 0.59]$	0.05 $[-0.59, 0.63]$
Geometry (exploratory)		
$\mathrm{AS}_{\mathrm{oop}}$ (out-of-plane only)	0.67 $[0.37, 0.93]$	0.67 $[0.35, 0.96]$
$\mathrm{AS}_{\mathrm{in}}$ (in-plane only)	0.67 $[0.33, 0.93]$	0.67 $[0.29, 1.00]$
SING's IS (B4)		
IS against row-space energy $1-\rho$	-0.55 $[-0.83, -0.10]$	-0.49 $[-0.84, 0.02]$

Table 8: Round-4 scores against SING's order (Kendall τ, percentile 95% CI, 2000 model resamples). A1: AS after removing $\sqrt{\rho_0/\rho}\,\Pi f$ (§4.1); ridge: our retrained translators on the test half (§4.4); $\mathrm{AS}_{\mathrm{cperm}}$ and Δ_{align}: §4.3; $\mathrm{NCM}_{\mathrm{null}}$, NZS and R: §4.5; geometry and IS: §B.4.

- *Null zero-shot accuracy* NZS: zero-shot classification from the translated null part alone, minus a permuted control. It passes its validity gate (no change under rescaling of Πf; τ=1 with planted content on all three bases), but it is not energy-free ($\tau(\cdot, \rho)$=0.45 and 0.53), and against SING it gives 0.30 $[-0.31, 0.89]$ (E12) and 0.38 (S11): inconclusive. Given ρ, its partial association with the order has negative point estimates, but every CI includes 0 (-0.60 $[-0.89, 0.50]$, -0.63 $[-0.92, 0.47]$ and -0.54 $[-0.93, 0.86]$ on all 13, E12 and S11).
- *Selectivity R*: AS after removing the top-20 between-class null directions (estimated on held-out images) over AS after removing the same energy at random. Within each model R differs from 1 (CI excludes 1 on 12 of 12 models), but in both directions (4 above 1, 8 below). Across models it does not reproduce the order (τ=0.15 and 0.05; k=5 and 50 give the same).

None of these scores tracks SING's order, and none is both energy-free and fully calibrated.

B.3 Organisation score (round 5, pre-registered)

The scores above measure how much class information the null part carries, which rises with its energy. $\mathrm{SO}_{\mathrm{null}}$ instead asks whether the null class means are *arranged* like the class texts. It is the Spearman correlation between a cross-validated (crossnobis) class-dissimilarity matrix of the null coordinates (Walther et al., 2016) and the dissimilarity matrix of SING's class-text embeddings, both double-centred, in the manner of representational similarity analysis (Kriegeskorte et al., 2008; Nili et al., 2014), divided by the square root of its split-half reliability. It needs no translator and is exactly invariant to rescaling the null part. A secondary score, XCC_{24}, is a cross-fitted alignment of the null and text class Gram matrices in random 24-dimensional frames. Both scores pass the pre-registered invariance, derangement and planted-code gates on all 13 models: exact invariance to scale and rotation, no signal after class means are deranged across classes, and a response to a semantic but not to an arbitrary planted class code. Both are estimable on 12 of E12 and 11 of S11.[1] After double-centring, the derangement gate is close to satisfied by construction. The semantic plant is built to raise a text-alignment score. These two gates are therefore sanity checks rather than evidence of validity (Appendix F). The corrected world C could not test $\mathrm{SO}_{\mathrm{null}}$: as ρ grows at

[1]Because $\mathrm{SO}_{\mathrm{null}}$ is scale-free, it is defined even where the null energy is negligible: DenseNet121 (ρ=1.8 $\times$ 10^{-8}) has $\mathrm{SO}_{\mathrm{null}}$=0.115. DenseNet121 is in neither E12 nor S11.

fixed L, its reliability falls below the floor of 0.10, so only one base has three estimable rows, and by the pre-registered rule this is not a kill. In worlds M and D both scores stay flat as L varies (within a factor 1.07 and 1.06). Against SING's order $\mathrm{SO}_{\mathrm{null}}$ gives τ=0.33 $[-0.29, 0.79]$ on E12 and 0.45 $[-0.21, 0.88]$ on S11: INCONCLUSIVE. It also follows ρ (0.42, 0.53), and given ρ its partial association is 0.03 and 0.23, with CIs that include 0. Without the reliability correction, raw SO gives 0.48 and 0.64 against SING and follows ρ more closely (0.58, 0.71; post hoc). XCC_{24}, averaged over random 24-dimensional frames, gives -0.12 and -0.05 (DOES-NOT-TRACK for this variant), although across all 13 models it agrees with $\mathrm{SO}_{\mathrm{null}}$ (τ=0.49). In post-hoc checks the full-null XCC gives 0.33 and 0.42 against SING, so at n=13 the organisation result depends on the method. The other pre-registered descriptives are in Appendix F. $\mathrm{SO}_{\mathrm{null}}$ is nearly unrelated to the between-class share η^2 (τ=0.03, 0.05). With the row-space class Gram projected out, the full-null XCC is at or below 0.05 on 7 of 13 models and above it on 6 (up to 0.43). This is below the pre-registered 75% bar for INHERITED. The plant check passed narrowly (minimum 0.109 against a bar of 0.10), so the pre-registered statement is NOT INHERITED. On about half of the models the congruence is therefore largely explained by the row-space class geometry, and we do not claim that null congruence is in general independent of it. The thresholds were set after scratch runs on five models for SO-type and three for XCC-type scores, which the pre-registration discloses. Appendix F reports how each label depends on its bar.

B.4 How AS grows with energy

Decomposition. Split the removed translated null component $\delta = A\Pi f$ into its part in the plane of $T(f)$ and the class text t and the orthogonal, out-of-plane part. The out-of-plane part changes the angle to t only at second order, by $\frac{1}{2}\cot\theta\,\|\delta_{\mathrm{oop}}\|^2/\|T(f)\|^2$. It does so whatever direction δ_{oop} has, so it is blind to the class text. If the translator's squared gain per null PC scales with the PC's variance as λ^s, a first-order argument predicts the energy exponent $a_{\mathrm{pred}} = (1+s)/2$.

Pre-registered tests. On a synthetic model with a known gain slope $s \in \{-0.4, \ldots, 0.2\}$ the fitted exponent is within 0.027 of $(1+s)/2$ (G3, pass). On the two true DINO backbones of §6, which were not used to design the test, the pre-registered checks pass on both: the Gaussian closed form orders the surrogate AS perfectly (Spearman 1), the out-of-plane-only removal shifts the angle ($[0.265, 0.269]$ and $[0.516, 0.523]$ degrees), the closed form is within 0.033 and 0.066 of the measured out-of-plane shift, and a_{pred} is within 0.1 of a_{fit} (0.603 against 0.669; 0.569 against 0.580; Table 18 in Appendix C). The out-of-plane share of AS is 0.52 and 0.80. These results are consistent with the account, but they are not a strong test of it. The Spearman check in G2 uses only four ρ_t points, and the G3 generator builds in the gain law. This gives a mechanism by which AS responds to null energy without regard to the class text. It does not show that the null space holds no semantics.

Exploratory, 13 models (post hoc). With the released translators the out-of-plane share of AS runs from 0.20 to 0.98, and the in-plane and out-of-plane parts track SING equally (τ=0.67 and 0.67 on E12). a_{pred} is within 0.1 of a_{fit} on 6 of the 8 models where both can be computed; it misses on BiT (0.08 against 0.40) and on DINOv2-L (0.29 against 0.39).

SING's IS. SING's invariant-space shift IS is not a proxy for row-space energy: $\tau(\mathrm{IS}, 1-\rho)$ is -0.62, -0.55 and -0.49 on the three sets, short of the pre-registered 0.6 on two. IS does, however, rise with null energy (CIs $[-0.89, -0.21]$, $[-0.83, -0.10]$ and $[-0.84, 0.02]$, excluding 0 on two sets), and it tracks SING's order (τ=0.74, 0.70 and 0.71).

B.5 True DINO backbones

In round 4 we evaluated the two DINO backbones that were released with a linear ImageNet head, ViT-B/16 and ViT-S/16 (Caron et al., 2021), together with a supervised timm AugReg ViT-B/16 as a control (Table 9). Their 50k top-1 is 78.06%, 76.91% and 84.54% (the DINO values are within 0.2 points of the published linear-evaluation accuracies), and the pre-registered feature checks pass. Each DINO head leaves a null space of dimension 536 (ρ=0.189 and 0.437); the AugReg head has none. There is no released SING

Model	top-1 (50k)	d_{null}	ρ	fidelity	AS (ridge) [rank/15]	95% PI from E12 law	NCM_{null}
DINO ViT-B/16	78.06	536	0.189	0.819	0.514 [8]	$[0.380, 0.884]$	56.9
DINO ViT-S/16	76.91	536	0.437	0.816	0.652 [9]	$[0.661, 1.573]$ (below)	53.4
AugReg ViT-B/16	84.54	0	0	–	–	–	–

Table 9: True DINO backbones with their released linear heads (§6), ridge translators on the test half. Rank among 15 models (1 = least leakage); prediction interval from the E12 energy law $\log \text{AS} \sim \log \rho$. AugReg ViT-B/16: supervised control with an empty null space.

translator for these models, so we use ridge translators fit as in §4.4 (fidelity 0.819 and 0.816) and compare with the 13 models under their ridge translators. AS is 0.514° (ViT-B/16) and 0.652° (ViT-S/16), ranks 8 and 9 of 15 (median of the 13: 0.399°). The pre-registered raw-AS question (Q-DN1) asks whether both DINO models rank among the four lowest. Its outcome is "SING-inconsistent", since both lie above the median of the 13. Raw ridge AS is not SING's own measure, however. It also tracks null energy, which for both models is above the median of the 13 (ρ=0.189 and 0.437). Post hoc, we therefore put more weight on the energy-adjusted reading (Q-DN2). Against the energy law fitted on E12, ViT-B/16 lies inside the 95% prediction interval ($[0.38, 0.88]$, predicted 0.58) and ViT-S/16 lies just below it ($[0.66, 1.57]$, predicted 1.02). ViT-S/16 thus has less AS than its energy predicts. That leans toward SING's account for that model, and the pre-registered outcome is mixed. For Δ_{align} and NCM_{null}, both models fall inside the prediction intervals from ρ. The pre-registered "least leaking" label, which needs both models among the four lowest, holds for neither score. ViT-S/16 alone is among the four lowest on both (Δ_{align} rank 3 of 15, NCM_{null} rank 2 of 14), while ViT-B/16 ranks 13 and 6. Both DINO heads are linear probes on frozen, concatenated block features, a different regime from the supervised heads of the 13 models, so this comparison is indicative only.

B.6 Size-corrected scores and class-level claims

Size-corrected scores. $\text{AS}_{\text{fair}} = \text{AS}_{\text{null}}/\text{AS}_{\text{rand}}$ divides AS by the AS of an energy-matched random removal. We used three families of controls, and name the family each time. In *family A*, the random subspace is Haar over the full feature space and changes the logits. *Family B* draws from the lowest-σ row-space directions of W. It is not logit-preserving either: it moves the logits by up to 14–74 logit SDs on models with a large span (VGG, ResNeXt; Table 11). *Family C* compares a 24-dimensional null subspace with a family-B control of matched energy. Point τ against SING on all 13 models is -0.18 (A), -0.41 (B) and -0.13 (C), and on S11 it is -0.56, -0.60 and -0.27. All fall below the pre-registered 0.3 bar.

Over eight normalisers (AS/ρ^a for $a \in \{0.25, 0.43, 0.5, 1\}$, $\text{AS}/d_{\text{null}}$, families A–C), the 95% CI excludes AS_{SING}'s 0.82 for 7 of 8 normalisers on all 13 models. On S11 it does so for exactly 5 of 8, so the weak "differs from" criterion passes at the threshold (5/8). The pre-registered reversal criterion (CI entirely below 0 under at least 5 of 8 normalisers) fails. It fails on all 13 models (0/8), and on S11 only family B meets it ($[-0.89, -0.23]$). That single pass comes from a control that is not logit-preserving. Normalising by the fitted exponent leaves a positive τ ($\rho^{0.43}$: +0.33 $[-0.15, 0.77]$; $\sqrt{\rho}$: +0.26), and family-A AS_{fair} over-corrects: its Spearman correlation with ρ is -0.79 $[-1, -0.23]$ on S11 (Fig. 5, Appendix D). Size-corrected rankings therefore *differ* from SING's, but the direction depends on the normaliser, and we make no reversal claim. Model ranks from AS_{SING} and AS_{fair} also shift with the null tolerance. Adjacent-tolerance τ falls to 0.87 and 0.82, below the pre-registered 0.9, while ρ stays at or above 0.92.

Class-level claims. For each of SING's testable claims we define a signed effect (positive = SING's direction) and bootstrap images within class (1000×, BCa intervals; Table 10). A claim *holds* if the CI is above 0, is *reversed* (rev.) if below, and is *not confirmed* (n.c.) if it covers 0. Rows that share a computation are merged. Under AS_{SING}, 7 claims hold, 3 reverse (one borderline) and 1 is contradicted. The reversed claims are C4 (DinoViT does not have the lowest class AS; ConvNeXt and DenseNet are lower), C8 (ResNet's porcupine and sports-car classes are not in its top decile) and C10 (jellyfish ≫ camel; effect -0.107, BCa $[-0.210, -0.006]$). C10 is borderline: its percentile interval $[-0.206, 7.9\times10^{-4}]$ covers 0. C9 ("no correlation" between ResNet and DinoViT class AS) is contradicted by a small correlation (-0.033, BCa CI excludes 0).

ID	Claim (short)	Effect AS_{SING} [BCa 95% CI]	AS_{SING}	AS_{fair}
C1/C11	DinoViT best ratio / least steerable (proxy)	$0.149\ [0.142, 0.156]$	holds	opp.
C2	ResNeXt worst ratio (proxy)	$0.323\ [0.303, 0.343]$	holds	opp.
C3/C6m	ResNet > DinoViT (class mean)	$1.383\ [1.372, 1.394]$	holds	opp.
C4	DinoViT least leakage	$-0.293\ [-0.298, -0.289]$	**rev.**	opp.*
C5	ResNeXt top-2 of 5	$1.442\ [1.424, 1.462]$	holds	opp.
C6 sd	ResNet class SD > DinoViT	$0.744\ [0.711, 0.785]$	holds	opp.
C7	DinoViT per-class $\lvert AS\rvert<1°$	$0.500\ [0.499, 0.500]$	holds	n/a
C8 porcupine	Porcupine $\geq$90th pct (ResNet)	$-5.2\ [-8.4, -1.9]$	**rev.**	opp.
C8 sports car	Sports car $\geq$90th pct (ResNet)	$-29.0\ [-34.1, -23.9]$	**rev.**	opp.
C9 spearman	No ResNet–DinoViT class corr.	$-0.033\ [-0.074, -0.012]$	contra.	contra.
C10	Jellyfish $\gg$ camel	$-0.107\ [-0.210, -0.006]$	**rev.**†	opp.
C12	Supp. Fig. 12b order (τ)	$0.821\ [0.56, 1.00]$‡	holds	n.c.‡
C13	EVA02 target-space order	–	untestable	untestable

Table 10: SING's class-level claims under AS_{SING} (BCa 95% CI, images resampled within class, so the CIs are conditional on the 13 fitted translators) and family-A AS_{fair}. rev. = CI on the opposite side of 0; n.c. = not confirmed; opp. = opposite side under AS_{fair}, a diagnostic since AS_{fair} over-corrects; contra. = a nonzero correlation where SING reports none. C1/C11 and C3/C6 mean are one computation each. C1/C2/C11 use $|AS|$ proxies for SING's ratio and steerability. *Under AS_{fair}, C4 coincides numerically with C1/C11 (same model attains the minimum) and is counted once. C6 SD: classes resampled. †Borderline: the percentile interval covers 0. ‡Model bootstrap; under AS_{fair}, $\tau=-0.18\ [-0.78, 0.35]$, not confirmed. C7 has no AS_{fair} analogue.

These differences may reflect protocol differences (for C10, SING used about 30 concepts and 1300 images). They are not caused by size correction. Under AS_{fair} (family A) the other testable effects fall on the opposite side (opp.). Because AS_{fair} over-corrects, we treat these as diagnostics, not reversals. The image-bootstrap CIs ignore uncertainty from the choice of models and normaliser, so they are narrow. For C6 (class SD) the bootstrap now resamples classes, since the effect is a spread over classes; the point estimate lies inside both the BCa and the percentile interval under both scores (AS_{SING}: $0.744\ [0.711, 0.785]$). Under AS_{fair} the effect is $-0.203\ [-0.243, -0.166]$, but correcting the SDs for within-class sampling noise shrinks it to -0.025 (AS_{SING}: 0.752), so the AS_{fair} diagnostic for C6 is fragile.

C Additional tables

Model	d_{null}	ρ	κ	γ	AS_{SING}	A1	AS_{surr}	fair A	fair B	B max Δlogit/SD
VGG16	3096	0.350	0.119	0.34	1.088	0.084	0.922	0.708	0.867	14.4
VGG19	3096	0.339	0.123	0.36	1.256	0.094	0.879	0.716	1.087	13.7
DenseNet121*	27	1.8×10^{-8}	2.5×10^{-8}	1.36	2.5×10^{-4}	–	2.4×10^{-4}	0.152	0.671	1.5×10^{-8}
ResNet50	1048	0.146	0.281	1.93	1.676	0.160	1.176	0.855	0.989	0.3
"DinoViT"	24	0.005	0.013	2.46	0.294	0.284	0.296	1.632	1.220	5.0×10^{-3}
EffNet-B0	280	0.130	0.030	0.23	0.443	0.076	0.378	0.889	1.267	0.8
BiT-R50x1*	1048	0.046	0.061	1.31	0.557	0.153	0.589	0.042	1.142	9.3×10^{-2}
ResNeXt101	1048	0.451	0.219	0.49	1.999	0.112	1.476	0.707	0.591	74.4
ConvNeXt-B	24	0.013	0.002	0.17	0.080	0.049	0.085	0.974	0.961	4.1×10^{-2}
Swin-L	536	0.274	0.151	0.55	1.019	0.105	1.170	0.829	1.130	7.5
DINOv2-L	1048	0.322	0.165	0.51	1.499	0.104	0.960	0.712	0.961	6.0
DeiT3-L	24	0.348	1.085	3.11	5.048	0.215	3.364	1.065	0.776	16.4
EVA02-L	24	0.009	0.031	3.47	0.417	0.302	0.411	2.612	1.306	2.3×10^{-2}

Table 11: Per-model scores on the 10k subset at tolerance 10^{-5}. κ: CLIP-space null energy; $\gamma = \kappa/\rho$; A1: AS after removing $\sqrt{\rho_0/\rho}\,\Pi f$; AS_{surr}: AS with leak-free surrogate null content; fair A/B: AS_{fair} families; last column: largest logit change of the family-B control in logit SDs (family B is not logit-preserving). *Excluded by the screen. Family C: Table 14.

Model	AS at α=0, ρ_t = 0.01	0.05	0.2	0.45	slope [95% CI]	AS(α)/AS(0), ρ_t=0.2 α=0.1	α=0.3
ResNet50	0.371	0.665	0.817	1.935	0.377 [0.375, 0.379]	1.088	1.508
ResNeXt101	0.074	0.175	0.453	1.105	0.690 [0.689, 0.692]	1.040	1.353
DINOv2-L	0.082	0.192	0.485	1.151	0.677 [0.676, 0.678]	1.049	1.363

Table 12: Leak-free null content (§4.2). AS at α=0 against the surrogate's null energy ρ_t; OLS slope of logAS on $\log \rho_t$ with 1000× image bootstrap; AS response to a planted concept at ρ_t=0.2.

Model	d=24, $\hat{a}$ = 0.431	0.388	0.485	d=100, $\hat{a}$ = 0.431	0.388	0.485
ResNet50	1.50 [1.48, 1.51]	1.75 [1.73, 1.77]	1.23 [1.21, 1.24]	1.18 [1.17, 1.19]	1.30 [1.28, 1.31]	1.05 [1.03, 1.06]
ResNeXt101	0.74 [0.73, 0.76]	0.85 [0.83, 0.87]	0.63 [0.61, 0.64]	0.97 [0.96, 0.99]	1.06 [1.04, 1.07]	0.87 [0.86, 0.89]
DINOv2-L	0.98 [0.97, 0.99]	1.06 [1.04, 1.07]	0.89 [0.88, 0.90]	0.91 [0.89, 0.92]	0.96 [0.95, 0.97]	0.85 [0.83, 0.86]
BiT-R50x1*	0.009 [0.009, 0.009]	0.010 [0.010, 0.010]	0.008 [0.008, 0.008]	0.012 [0.012, 0.012]	0.013 [0.013, 0.013]	0.011 [0.011, 0.011]

Table 13: PC bands against the fitted energy law (§B.1): H with 1000× image-bootstrap CIs at three values of $\hat{a}$ (E1 point estimate and CI ends). $H < 1$: top PCs carry less AS than energy predicts. *BiT is reported but not counted.

	VGG16	VGG19	Dense	RN50	DinoViT	Eff	BiT	RNX	ConvNeXt	Swin	DINOv2	DeiT3	EVA02
fair C	1.32	1.35	0.71	1.14	1.20	1.44	135.6	1.04	0.97	1.08	1.00	0.79	1.33

Table 14: Family-C $\mathrm{AS}_{\mathrm{fair}}$ per model (`r2/results/fix2_models.csv`).

Kind	Null content	all 13	E12	S11	partial r given ρ (S11)
content-free	$\mathcal{N}(\mu, \Sigma)$ (round 3, $\mathrm{AS}_{\mathrm{surr}}$)	0.69 [0.39, 0.92]	0.64 [0.31, 0.90]	0.64 [0.27, 0.92]	0.52 [−0.07, 0.94]
content-free	$\mathcal{N}(0, \Sigma)$	0.72 [0.43, 0.94]	0.67 [0.38, 0.93]	0.67 [0.35, 0.96]	–
content-free	isotropic $\mathcal{N}(0, I)$	0.49 [0.04, 0.89]	0.39 [−0.04, 0.87]	0.60 [0.16, 0.96]	–
content-free	mean only (μ)	0.49 [0.07, 0.83]	0.39 [−0.05, 0.74]	0.38 [−0.16, 0.79]	–
content-free	real c permuted across images, $\mathrm{AS}_{\mathrm{perm}}$	0.69 [0.39, 0.92]	0.64 [0.31, 0.90]	0.64 [0.27, 0.92]	0.52 [−0.07, 0.94]
class content	real c permuted within class	0.79 [0.50, 1.00]	0.76 [0.46, 0.97]	0.78 [0.44, 1.00]	–
excess	$X = \mathrm{AS}_{\mathrm{SING}}/\mathrm{AS}_{\mathrm{surr}}$	0.54 [0.14, 0.83]	0.58 [0.21, 0.86]	0.60 [0.24, 0.91]	0.79 [0.02, 0.97]
excess	$\mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{surr}}$	0.59 [0.15, 0.89]	0.64 [0.24, 0.90]	0.67 [0.27, 0.96]	0.81 [0.02, 0.98]
excess	$\mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{perm}}$	0.62 [0.15, 0.92]	0.67 [0.25, 0.96]	0.71 [0.28, 1.00]	0.77 [0.04, 1.00]
excess	within-class/across-image	0.41 [0.00, 0.69]	0.45 [0.12, 0.74]	0.45 [0.09, 0.75]	0.57 [−0.48, 0.93]
–	real ($\mathrm{AS}_{\mathrm{SING}}$)	0.82 [0.56, 1.00]	0.79 [0.51, 1.00]	0.82 [0.51, 1.00]	0.78

Table 15: Surrogate null content (§4.2): Kendall τ against SING [percentile 95% CI], released translators, 3 draws per surrogate (5 for the Gaussian), each rescaled to the model's own null energy. Last column: partial Spearman given ρ on S11 with its CI.

Model	own ρ	AS(α)/AS(0), ρ_t=0.2 α=0.1	α=0.3	AS(α)/AS(0), own ρ α=0.1	α=0.3	AS (°) own ρ, α=0/0.3	r_E
ResNet50	0.146	1.088	1.508	1.099	1.581	0.768/1.214	2.39
ResNeXt101	0.451	1.040	1.353	1.018	1.137	1.107/1.259	5.37
DINOv2-L	0.322	1.049	1.363	1.019	1.169	0.762/0.891	4.92

Table 16: Planted content (§4.2): AS response to a planted concept at the surrogate energy ρ_t=0.2 and at each model's own ρ, absolute AS at its own ρ, and $r_E = \mathrm{AS}(\rho_{75})/\mathrm{AS}(\rho_{25})$, the AS change the energy law gives across the interquartile range of ρ.

Estimator	World C: $\lvert\tau(\cdot,\rho)\rvert \leq 0.2$ RN50	RNX	DINOv2	World M: $\tau(\cdot,L) \geq 0.8$ RN50	RNX	DINOv2	World D: $\tau(\cdot,L) \geq 0.6$ RN50	RNX	DINOv2	diagnostic
ρ	1.00	1.00	1.00	1.00	1.00	1.00	$-0.03^{\dagger}$	$0.00^{\dagger}$	$0.06^{\dagger}$	no
AS	0.97	1.00	1.00	0.33	1.00	1.00	0.12	1.00	1.00	no
A1	0.73	0.64	0.48	0.52	−0.76	−0.79	−1.00	0.70	0.61	no
$\mathrm{AS}_{\mathrm{gauss}}$	1.00	1.00	1.00	0.94	1.00	1.00	−0.52	−1.00	−0.76	no
X	−0.03	−0.88	−0.52	−0.45	−0.73	−0.76	0.91	1.00	1.00	no
Δ_{align}	−1.00	−1.00	−1.00	−0.21	0.58	0.12	0.91	1.00	1.00	no
$\mathrm{NCM}_{\mathrm{null}}$	−0.72	−0.99	−0.99	0.81	0.88	0.44	1.00	1.00	0.99	no
selectivity R	−0.91	−0.21	−0.15	−0.06	−0.79	0.70	−0.33	−0.94	−0.33	no
exponent a	0.59	0.47	0.43	0.31	0.53	0.47		–		

Table 17: Calibration worlds (§4.5): Kendall τ against ρ in world C and against L (class-mean null energy as a share of total feature energy) in worlds M and D, on three bases. Criteria in the header; "diagnostic" requires all nine. †In world D, ρ is fixed by design and varies only through the small cross term between class means and permuted residuals, so its τ with L is ≈ 0 (near-ties) and carries no information. Last row: fitted energy exponent.

Model	AS	$\mathrm{AS}_{\mathrm{in}}$	$\mathrm{AS}_{\mathrm{oop}}$	oop share	oop closed-form err.	$a_{\mathrm{pred}} = (1+s)/2$	a_{fit}
Exploratory (post hoc): 13 models, released translators							
VGG16	1.088	0.786	0.667	0.61	0.086	0.42	0.36
VGG19	1.256	0.857	0.678	0.54	0.086	0.42	0.36
ResNet50	1.676	0.938	1.645	0.98	0.160	0.40	0.44
"DinoViT"	0.294	0.291	0.096	0.33	0.016	0.33	–
EffNet-B0	0.443	0.395	0.201	0.45	0.024	0.69	0.67
BiT-R50x1	0.557	0.502	0.405	0.73	0.052	0.08	0.40
ResNeXt101	1.999	1.169	1.287	0.64	0.116	0.44	0.45
ConvNeXt-B	0.080	0.081	0.016	0.20	0.002	0.79	–
Swin-L	1.019	0.938	0.849	0.83	0.096	0.41	0.45
DINOv2-L	1.499	0.857	1.096	0.73	0.124	0.29	0.39
DeiT3-L	5.048	1.926	4.626	0.92	0.185	0.37	–
EVA02-L	0.417	0.393	0.229	0.55	0.044	−0.03	–
Confirmatory (pre-registered G2): true DINO, ridge translators							
DINO ViT-B/16	0.514	0.384	0.267	0.52	0.033	0.60	0.67
DINO ViT-S/16	0.652	0.630	0.519	0.80	0.066	0.57	0.58
Synthetic calibration (G3): known gain slope s							
s	−0.4, −0.2, 0.0, 0.2				$a_{\mathrm{fit}} - (1+s)/2$:	−0.002, −0.004, 0.006, 0.027	

Table 18: Geometric decomposition of AS (§B.4), degrees. oop: out-of-plane; closed-form error: relative error of $\frac{1}{2}\cot\theta\|\delta_{\mathrm{oop}}\|^2/\|T(f)\|^2$ among images with $\|\delta_{\mathrm{oop}}\|/\|T(f)\| < 0.5$; a_{fit}: energy exponent fitted on matched-d cells (– where the cell fit is not available). DenseNet omitted ($\rho \approx 0$).

D Energy law, Exp 1 and checkpoint accuracy

Model	r	2σ band	R^2 (vs $d \equiv$ vs ρ_d)
ResNet50	10.15	[5.83, 14.48]	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512

Table 19: Exp 1: $r = \mathrm{AS}(d{=}1048)/\mathrm{AS}(d{=}24)$ over random null subspaces (20 seeds). For random subspaces ρ_d is linear in d, so the two R^2 coincide. At matched $d{=}24$ across 8 models, τ against SING is 0.29 ($p{=}0.40$).

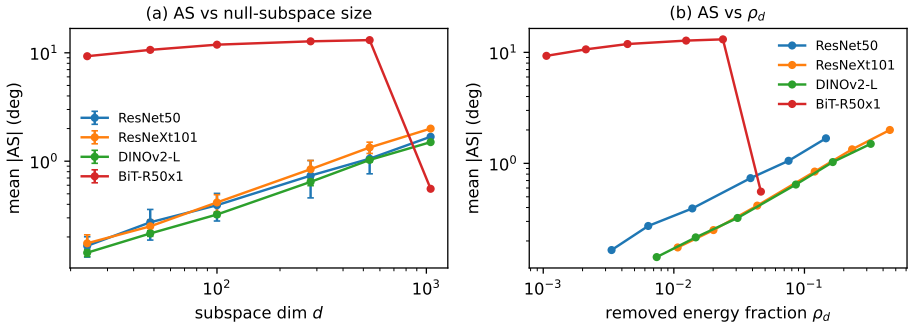


Figure 3: Exp 1: mean |AS| against subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation). Full-null anchors: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°.

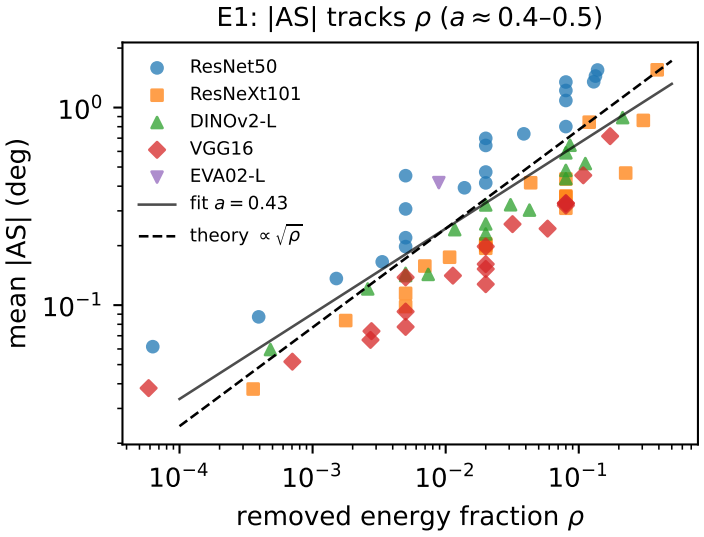


Figure 4: Energy law: |AS| against ρ over 76 cells on five models (ResNet50, ResNeXt101, DINOv2-L, VGG16, EVA02-L) at fixed $d \in \{24, 100, 280\}$ and fixed $\rho \in \{0.5\%, 2\%, 8\%\}$. Solid: pooled fit ($a$=0.431); dashed: $\sqrt{\rho}$. At fixed d the high/low-energy ratio exceeds 3× on all 4 evaluable models; at fixed ρ the max/min across d exceeds 1.5 in several cells (up to 2.29), so the pre-registered rating is *mixed*.

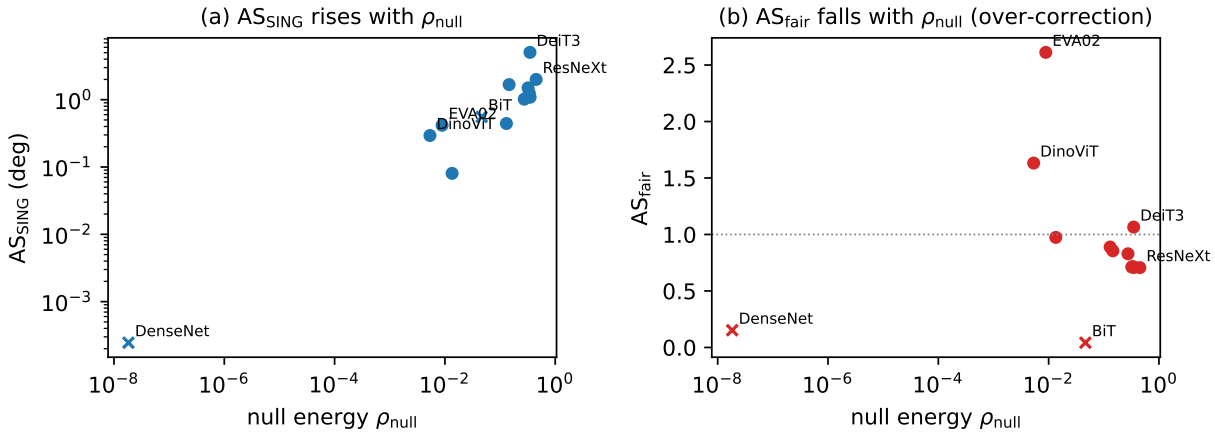


Figure 5: (a) AS_{SING} rises with ρ. (b) Family-A AS_{fair} falls with ρ: it over-corrects. $\times$: models excluded by the screen.

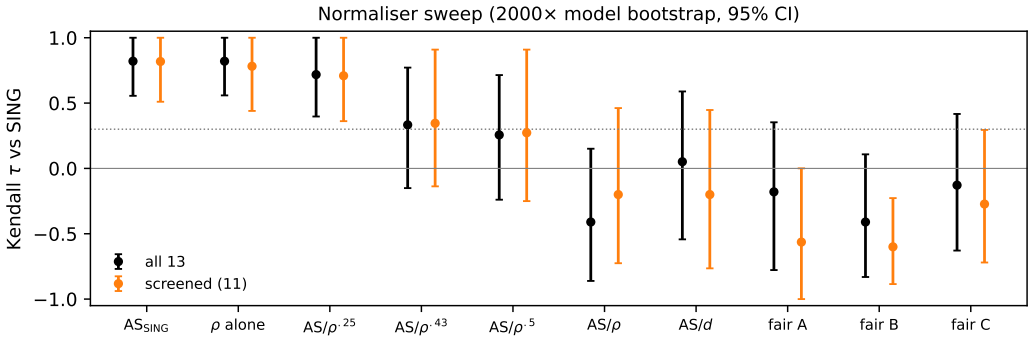


Figure 6: τ against SING for each normaliser and for ρ alone (2000× model bootstrap), on all 13 models and on S11. Dotted line: the τ=0.3 bar.

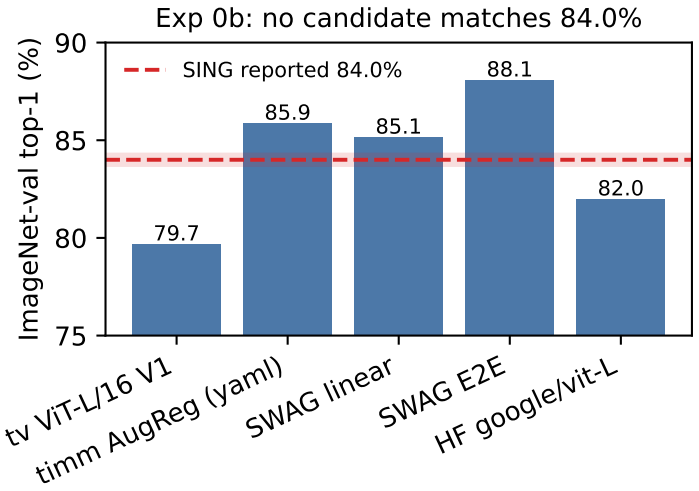


Figure 7: Top-1 on ImageNet-val (native evaluation transform) of the five public ViT-L/16 candidates. None matches SING's reported 84.0%.

E Planted leakage, score search and the flatness rule

Planted leakage (E3). We planted a CLIP concept direction (`red`, `striped`, `water`) into the null spaces of 8 models at log-uniform $\alpha \in [0.01, 1]$ over 50 trials, with logits unchanged (max $|\Delta\text{logit}| \leq 2.3\times10^{-5}$). Kendall τ against the planted ranking, averaged over concepts, is 0.386 $[0.324, 0.448]$ for AS_{SING} and 0.109 $[0.046, 0.171]$ for AS_{fair} (both α=0-subtracted). The pre-registered target ($\text{AS}_{\text{fair}} \geq 0.8$, $\text{AS}_{\text{SING}} \leq 0.5$) was not met. Neither score recovers planted strength well across models.

E8 (pre-registered; killed). A probe-based null-content score (ridge-probe R^2 on null coordinates, z-scored against energy-matched random subspaces) gave $\tau{=}-0.331$ $[-0.392, -0.273]$ on the planted task and was flat on 0/4 models. Three implementation caveats apply: the z-score saturates, λ was chosen by approximate leave-one-out rather than the pre-registered exact cross-validation, and λ sat at the grid floor in 18 of 96 fits.

E9 (pre-registered; killed). We screened 20 candidate scores on two development concepts. The rule required $\tau \geq 0.8$ and flatness (max / min ≤ 1.5 across nested null subspaces) on at least 3 of 4 models at $\alpha \in \{0.1, 1\}$. No candidate passed. The best was M1, a debiased null-leakage estimate $\hat{L}^2 = (\|\sum_i Pf_i s_i\|^2 - \sum_i \|Pf_i\|^2 s_i^2)/(n(n{-}1)\mathbb{E}\|f\|^2)$ with standardised concept scores s_i, scored as $\sqrt{\max(\hat{L}^2, 0)}$. It reached $\tau{=}0.976$ $[0.963, 0.988]$ but was flat on 0/4 models.

The flatness rule is ill-posed. We tested the rule on synthetic Gaussian features with a known head ($m{=}400$, $d_{\text{null}}{=}300$). Our exact oracle is the planted amplitude inside the removed subspace. It fails the rule with max / min 3.55 and 3.48 ($\approx \sqrt{300/24}$), because a larger removed subspace contains more of the plant. A revised rule (max / min of score/oracle ≤ 1.5, post hoc) is passed by M1 (ratios 1.00–1.03). Under the revised rule M1 is flat on 4/4 models, and E8's score on 0/4. E8 and E9 remain killed as registered.

Holdout (post hoc). We ran the E9 holdout on four held-out concepts (water, green, thin, knitted; 50 trials at seed 1; linear and nonlinear plants) for the six candidates declared in the round-2 pre-registration plus AS_{SING} and AS_{fair}, with the E9 pass rule (Table 20). Only M1 meets both τ thresholds. Under the registered flatness rule no candidate passes; that rule also fails for the oracle (above). Under the revised rule, M1 passes. Paired over trials (1000×, seed 0), $\tau(\text{M1}){-}\tau(\text{AS}_{\text{fair}})$ is 0.77 [0.71, 0.82] for the linear plant and 1.01 [0.96, 1.06] for the nonlinear plant, so M1 tracks the planted leakage better than AS_{fair} under both

Score	lin. τ [95% CI]	nl. τ	flat	rev. flat	pass
M1	0.93 [0.91, 0.95]	0.97	0/0	4/4	rev.
M2	0.44 [0.39, 0.48]	0.46	0/0	0/0	no
S9	0.45 [0.40, 0.51]	0.44	0/1	2/0	no
S10	0.37 [0.33, 0.42]	0.52	3/4	0/0	no
M5 ($AS/\sqrt{\rho}$)	0.22 [0.18, 0.27]	0.15	0/0	0/0	no
S4 (AS/ρ)	−0.08 [−0.15, −0.01]	−0.14	0/0	0/0	no
AS_{SING} (raw)	0.33 [0.27, 0.40]	0.36	0/0	0/0	no
AS_{fair} (sub.)	0.17 [0.11, 0.22]	−0.04	0/0	0/0	no

Table 20: E9 holdout (post hoc; AS_{fair} is α=0-subtracted; water, green, thin, knitted; 50 trials, seed 1; 8 models). Flat = number of the 4 large-null models flat at α=0.1/1 under the registered rule; rev. flat = under the revised score/oracle rule. Pass: lin. $\tau \geq 0.8$, CI lower ≥ 0.7, nl. $\tau \geq 0.6$, flat on $\geq 3/4$ at both α (yes = registered rule, rev. = only with the revised rule).

plants. The revised rule also changes which candidate counts as flat: S10, the only candidate flat under the registered rule (3/4 and 4/4 models at α=0.1/1), is flat on 0/4 and 0/4 under the revised one. The plant is written along the null projection that M1 itself measures, and M1, unlike the AS-family scores, is given the concept score that defines the plant. This comparison therefore favours M1 by construction. One of the four held-out concepts, water, was also an E3 plant concept whose results we had seen before choosing the holdout set; its inclusion was pre-registered, but the set is not fully unseen. M1 was picked after the DEV results and the flatness rule was revised after the oracle test, so this holdout is post hoc. E9 stays a KILL as registered, and M1 is a candidate for a future pre-registered test, not a validated score.A reduced-draw mode (5 instead of 20 random draws for AS_{fair}, 2 instead of 5 nested bases, S9 omitted), declared before it was run, gives the same M1 and AS_{SING} values and the same flatness verdicts. AS_{fair} moves by +0.02 (linear) and +0.06 (nonlinear; 0.057), and the run takes 36 min on 8 CPU cores. It fails both targets we set ($\pm$0.03, under 30 min), so the full run (1.95 h) is the reference (Appendix H).

M1 on the 13 models (post hoc, descriptive). At α=0 on the full null, averaged over SING's 173 concepts, M1 gives τ=0.667 $[0.29, 0.94]$ against SING, an interval that includes AS_{SING}'s value. M1 orders the models much as ρ does (τ=0.79), as expected for a score scaled by total feature energy. Its holdout validation favours it by construction (the plant is written along the projection it measures), so this agreement is not independent evidence about energy. We treat M1 as a candidate for a future pre-registered test, not as a validated score.

F Round 5: calibration fix and organisation score

Calibration fix. Round 4 built the worlds of §4.5 from each image's own residual $r = c - \mu_y$. Within a class these residuals sum to zero, so a class mean estimated on one subset of images is the negative of the residual mean of the rest. This biases any estimator that compares class means across disjoint images. Round 5 permutes r across all images (seed 99000) before rescaling, and keeps the round-4 results file. The fix was coded and run, and the corrected world τ values for ResNet50 were seen, before `PREREG_R5.md` was pushed; the pre-registration states this. Permuting r across images keeps the class means, the pooled residual distribution and the total residual energy, but not class-specific residual covariance or the image-level coupling between row-space content and the null residual, so the worlds are cleaner than real features. 50 of 78 entries change (Table 17 against Table 24). By $|\Delta\tau|$ the largest changes (0.48 each) are X in world C on ResNet50 ($-0.52 \to -0.03$); A1 in world C on DINOv2-L ($0.00 \to 0.48$); X in world C on DINOv2-L ($-1.00 \to -0.52$). The most consequential is NCM_{null} in world M on DINOv2-L (from 0.80 to 0.44, so it now misses the world-M criterion on that base). In the corrected world C, NCM_{null} falls as energy grows at fixed content (τ from −0.99 to −0.72). In round 4 the world-M range of that score was 0.80–0.92. The content-diagnostic verdict is unchanged (none before, none after). In world D, ρ is no longer exactly constant, because the cross term between class means and permuted residuals is not exactly zero. Its τ with L is therefore now defined, and it is close to 0.

Scores. For $\mathrm{SO}_{\mathrm{null}}$, the images are split into four class-stratified folds (seed 99200). Crossnobis distances between class means are computed from folds $\{0,1\}$ and from folds $\{2,3\}$ (Walther et al., 2016). Each distance matrix and the text dissimilarity $1-\cos$ are double-centred. $\mathrm{SO}_{\mathrm{null}}$ is the mean Spearman correlation of the two halves with the text matrix, divided by the square root of their reliability r_{rel} (Spearman between halves), and is estimable if $r_{\mathrm{rel}} \geq 0.10$. We added double-centring after the scratch run. Without it, a class-mean derangement left SO near its real value on ResNet50 (disclosed in the pre-registration). XCC_{24} takes centred half-means from two image halves per class and forms their off-diagonal cross Gram, normalised against the centred text Gram and averaged over 50 Haar 24-frames (seeds 99100+s). It is estimable if its self-term $z \geq 5$ (10 class folds, seed 99600). Permutation nulls use 200 joint class permutations (seed 99300).

Gates and outcomes. G1 (exact invariance to $\beta \in \{0.1, 10\}$ and an orthogonal map, seed 99500): SO pass, XCC pass. G2 (class-mean derangement, seeds 94000+r): no failures; the largest mean derangement score is 0.002 (SO) and 0.001 (XCC). G3 (world S, seed 99400): on all 13 models, the semantic plant at α=0.3 raises SO by at least 13 null SDs, and the arbitrary plant changes it by at most -5.0 (it lowers it). G4: SO testable on one base, XCC on two (both pass where testable); not a kill. G5: estimable on all 13. No kill fired. Inheritance: F1-partial ≤ 0.05 on 7 of 13 models (54%, below the 75% bar), with the plant check passed (minimum 0.109), so the pre-registered statement is NOT INHERITED.

What the gates show. G2 is close to satisfied by construction once both matrices are double-centred. Deranging class means permutes the rows and columns of the class dissimilarity matrix, so the gate mainly guards against per-class residual spread, which is the problem the scratch run exposed. G3 plants a random projection of the text geometry. That plant is built to raise a text-alignment score, so this gate is a sensitivity check, not evidence of validity. The arbitrary plant lowers SO because it adds class structure unrelated to the texts. SO therefore measures how the null class means are arranged relative to the texts (relative alignment), not how much class information the null part holds.

Centring correction (post hoc). The committed double-centring subtracts the off-diagonal row means of each hollow (zero-diagonal) dissimilarity matrix. For an additive class effect a_i+a_j this leaves a residual of $(a_i+a_j-2\bar{a})/(N-1)$, and the row sums of the output are not centred, so it does not remove additive per-class effects exactly. In a synthetic test with no content and large class offsets correlated with the text marginal, this version gives a spurious, estimable $\mathrm{SO}_{\mathrm{null}}$. We reran $\mathrm{SO}_{\mathrm{null}}$ with an exact hollow double-centring (`r5/so_exact_r5.py`; same features, folds, text, permutation null and gates; the original code path reproduces the committed values to 2×10^{-16}). This is a separately labelled post-hoc rerun. The registered values and verdict are unchanged in the record. Per model, $\mathrm{SO}_{\mathrm{null}}$ changes by at most 0.0012. The order of the 13 models is unchanged (τ=1.00), 13 of 13 models stay estimable, G2 passes on 13 and G3 on 13, τ with SING's order is 0.38 on all 13 models (0.33/0.45 on E12/S11), and the outcome is still INCONCLUSIVE.

Sensitivity to the bars (post hoc). Two bars sit at the scratch values: the inheritance bar of 0.05 (scratch maximum 0.040) and the plant-check bar of 0.10 (scratch minimum 0.103, final minimum 0.109). The reliability floor of 0.10 was set after the scratch run showed ResNet50's world-C rows falling below it. With a plant bar of 0.11, the inheritance statement becomes UNINFORMATIVE. Raising the inheritance bar to 0.12 gives INHERITED (10 of 13 models). Across reliability floors from 0.005 to 0.4, SO keeps its label (INCONCLUSIVE), and no kill fires. With α=0.1 in G3, SO passes on 12 and XCC on 13 of 13 models. At floors of 0.005–0.05, G4 is testable for SO on two bases and passes on both (max/min ratio at most 1.04). On DINOv2-L, the base testable at the pre-registered floor, the rows cover only ρ=0.005–0.013. The row-space analogues give $\mathrm{SO}_{\mathrm{row}}$ 0.30 and 0.35 and $\mathrm{XCC}_{\mathrm{row}}$ 0.27 and 0.35 against SING. All numbers in this paragraph come from `r5/posthoc_r5.py`.

Model	ρ	η^2	SO_{null}	rel	XCC_{24}	z	G2 SO / XCC	G3 SO sem / arb	F1-partial
VGG16	0.35	0.24	0.310	0.41	0.098	21.7	1.1×10^{-4} / 4.6×10^{-4}	196 / −14.1	0.052
VGG19	0.34	0.26	0.288	0.44	0.090	23.1	−0.002 / 2.3×10^{-4}	202 / −13.3	0.029
DenseNet121	1.8×10^{-8}	0.21	0.115	0.26	0.123	21.8	-1.7×10^{-5} / 2.6×10^{-4}	170 / −36.2	−0.051
ResNet50	0.15	0.19	0.312	0.25	0.169	18.2	−0.002 / 8.3×10^{-4}	267 / −53.4	0.040
"DinoViT"	5.4×10^{-3}	0.13	0.114	0.14	0.122	30.7	-7.4×10^{-4} / 9.6×10^{-4}	29 / −12.6	0.031
EffNet-B0	0.13	0.25	0.202	0.36	0.120	27.9	0.002 / 0.001	148 / −20.3	0.162
BiT-R50x1	0.05	0.24	0.347	0.44	0.219	26.6	7.9×10^{-4} / 0.001	115 / −7.6	0.098
ResNeXt101	0.45	0.63	0.276	0.67	0.077	10.0	0.002 / 2.0×10^{-4}	146 / −15.9	0.002
ConvNeXt-B	0.01	0.42	0.160	0.57	0.167	45.2	-2.8×10^{-4} / -3.1×10^{-4}	52 / −18.9	0.119
Swin-L	0.27	0.30	0.488	0.63	0.218	28.5	7.9×10^{-4} / 2.1×10^{-4}	90 / −24.8	0.426
DINOv2-L	0.32	0.37	0.209	0.49	0.051	27.6	5.0×10^{-4} / 3.0×10^{-4}	225 / −17.8	0.001
DeiT3-L	0.35	0.76	0.523	0.90	0.559	18.7	-4.2×10^{-4} / 0.001	13 / −67.0	0.429
EVA02-L	8.9×10^{-3}	0.38	0.039	0.49	0.046	36.9	-5.5×10^{-5} / -1.0×10^{-4}	49 / −5.0	0.036
True DINO (values only)									
DINO ViT-B/16	0.19	0.29	0.302	0.42	0.163	31.7	–	–	−0.040
DINO ViT-S/16	0.44	0.31	0.440	0.53	0.275	31.8	–	–	−0.002

Table 21: Round-5 scores and gates per model. rel: split-half reliability of SO_{null}; z: XCC self-term; G2: mean over 5 class-mean derangements; G3: change at α=0.3 in null-SD units for the semantic / arbitrary plant; F1-partial: full-null XCC with the row-space class Gram projected out. True DINO models are values only.

Base	Score	G4: estimable rows, max / min	World B range	World M range	World D range
ResNet50	SO_{null}	0/12, not estimable	0.312–0.312 (12)	0.223–0.236 (11)	0.230–0.236 (4)
ResNet50	XCC_{24}	1/12, not estimable	0.169–0.169 (12)	0.122–0.128 (12)	0.122–0.127 (6)
ResNeXt101	SO_{null}	2/12, not estimable	0.276–0.276 (12)	0.248–0.265 (12)	0.252–0.265 (9)
ResNeXt101	XCC_{24}	3/12, 1.009 (pass)	0.077–0.077 (12)	0.072–0.073 (12)	0.072–0.074 (11)
DINOv2-L	SO_{null}	3/12, 1.009 (pass)	0.209–0.209 (12)	0.179–0.181 (12)	0.179–0.183 (11)
DINOv2-L	XCC_{24}	6/12, 1.066 (pass)	0.051–0.051 (12)	0.043–0.044 (12)	0.041–0.044 (12)

Table 22: Round-5 scores in the corrected worlds. G4 uses world C; worlds B (whole null part rescaled; truth: flat), M and D are descriptive. Ranges over estimable rows, count in parentheses.

Score	Set	n	τ vs SING [95% CI]	$\tau(\cdot,\rho)$	partial r given ρ
SO_{null}	E12	12	0.33 $[-0.29, 0.79]$	0.42	0.03 $[-0.67, 0.74]$
SO_{null}	S11	11	0.45 $[-0.21, 0.88]$	0.53	0.23 $[-0.41, 0.82]$
SO_{null}	all13	13	0.38 $[-0.11, 0.75]$	0.46	0.04 $[-0.62, 0.68]$
XCC_{24}	E12	12	−0.12 $[-0.77, 0.44]$	−0.03	0.09 $[-0.75, 0.75]$
XCC_{24}	S11	11	−0.05 $[-0.73, 0.54]$	0.02	0.14 $[-0.75, 0.79]$
XCC_{24}	all13	13	−0.13 $[-0.69, 0.39]$	−0.05	0.11 $[-0.65, 0.69]$

Table 23: Round-5 confirmatory test: Kendall τ with SING's order (model bootstrap, percentile 95% CI), τ with ρ, and the partial Spearman given ρ. Labels: SO_{null} INCONCLUSIVE, XCC_{24} DOES-NOT-TRACK.

Estimator	World C			World M			World D		
	RN50	RNX	DINOv2	RN50	RNX	DINOv2	RN50	RNX	DINOv2
ρ	1.00	1.00	1.00	1.00	1.00	1.00	–	–	–
AS	1.00	1.00	1.00	0.33	1.00	1.00	0.24	0.97	0.94
A1	0.70	0.55	0.00	0.45	−0.76	−0.85	−1.00	0.48	0.94
$\mathrm{AS}_{\mathrm{gauss}}$	1.00	1.00	1.00	0.94	1.00	1.00	−0.55	−1.00	−0.94
X	−0.52	−1.00	−1.00	−0.67	−0.76	−0.88	0.97	1.00	0.97
Δ_{align}	−1.00	−1.00	−1.00	−0.15	0.61	0.18	0.94	1.00	0.94
$\mathrm{NCM}_{\mathrm{null}}$	−0.32	−1.00	−1.00	0.92	0.88	0.80	0.93	1.00	1.00
selectivity R	−0.88	−0.45	−0.30	−0.06	−0.85	0.79	−0.73	−0.88	−0.48
exponent a	0.55	0.49	0.30	0.28	0.52	0.42		–	

Table 24: Round-4 calibration worlds with in-sample residuals, superseded by Table 17; kept for comparison.

G Round 6: energy statistics for published null-space and residual scores

Scores. ViM (Wang et al., 2022): origin $u{=}-W^{+}b$, uncentred covariance of $f{-}u$, residual space after the top D eigenvectors ($D{=}1000$ for $d \geq 2048$ and 512 otherwise, as in the official code), α from the fit half; model-level score is the mean virtual-logit probability (descriptive: mean of the official score s). ActSub $S_{\rightarrow}$ (Zöngür et al., 2025): full SVD of W, split index from the official rule, a random 10% reference set of the fit half, 10 nearest cosine neighbours, $S_{\rightarrow}{=}-\log(1-\bar{c})$. CORE S_{mem} (Yang et al., 2026): cosine between the residual of z with respect to the predicted class's weight and that class's normalised mean residual on the fit half (descriptive: the same on Πf). Conditional $\mathcal{V}$-information (Hewitt et al., 2021): held-out NLL gain of a softmax probe on [a scaled copy of the logits; the top 64 principal coordinates of Πf] over the logits alone, with an ℓ_2 penalty chosen once per model by inner cross-validation and then frozen (descriptive: top-1 correctness as target). The NuSA-style reference is the mean of $\|\Pi f\|/\|f\|$. Every image is scored by quantities fitted on the other half. We checked ViM's residual dimension and ActSub's split rule and neighbour count against the official code. The main deviation is that we have no training features, so every fitted quantity uses 5 validation images per class from the other half of a class-stratified split. The scores are therefore surrogates of the published ones, with three known fidelity limits: CORE's per-class mean residuals rest on 5 images, fewer than the about 13 per class it needs; ActSub's reference set is 10% of this fit half, not of a training set; and $\mathcal{V}$-information uses a restricted probe family, since a full map of the 1000 logits is not identifiable from 5 images per class. We did not run CANARY (Shin & Kim, 2026), whose per-class $k{=}100$ neighbour rule needs more references per class, nor NECO (Ben Ammar et al., 2024) or the nearest-neighbour score of Sun et al. (2022).

Pre-registered verdicts. Table 25 gives the registered outcomes. All four scores are estimable on enough models ($\mathcal{V}$-information on 11 of E12 and 10 of S11), so $n{=}4$. ViM is NOT ENERGY-DRIVEN ($\tau_{\mathrm{EQ}}{=}0.88$ and 0.89, median $R_B{=}0.06$). ActSub and CORE keep their order (0.79/0.78 and 0.85/0.82) with median R_B 0.14 and 0.11, just above the 0.1 bar, so both are MIXED. $\mathcal{V}$-information ($\tau_{\mathrm{EQ}}{=}0.20$ and 0.33, $R_B{=}0.19$) is MIXED. The counting rule gives $k{=}0$ of $n{=}4$, for which the pre-registration states that the confound is largely specific to SING. We withdraw that statement: the post hoc positive controls of §5 show that the rule does not flag SING's own AS, so k carries no information about other scores, and we do not generalise from it. The registered content labels (PARTIAL for ViM, ActSub and CORE; CONTENT-RESPONSIVE for $\mathcal{V}$-information) are listed in Table 25 and the ledger as registered outcomes only. We do not use them as readings, for the reasons below. The registered statement that null content is measurable needs two scores that are both not energy-driven and content-responsive, and it is not made.

Content tests and the C2 flag. C1 permutes Πf across images, and C3 adds an arbitrary class code ($\alpha{=}0.3$), both in image-bootstrap SDs. The within-class permutation C2 moves every score by more than 1 SD on more than 3 models (ViM 8, ActSub 11, CORE 9, $\mathcal{V}$-information 12 of 13), so the pre-registered ESTIMATOR-NOISY flag is set for every score. That flag assumed that C2 preserves the content a score

measures. It need not. In the synthetic check of `r6/tests/test_posthoc_r6.py` (post hoc), the null part carries the image-level noise of the logits, so logits and null part together identify the class. The conditional $\mathcal{V}$-information gain is 1.18 nats, and after a within-class permutation, which leaves every class's null distribution exactly unchanged, it is -9.7×10^{-3}. The C2 flag therefore cannot separate a real loss of within-class or image-level information from estimator noise. Permuting Πf across images in C1 also breaks the coupling between row-space and null parts, and the bootstrap SDs over 10 000 images are small, so C1 and C3 effects mix content with that coupling (the NuSA share passes C1 on 9 models). For $\mathcal{V}$-information, |C2| exceeds |C1| on 11 of 13 models. The semantic class code (descriptive) moves the scores in the same direction as the arbitrary code on most models. None of the four scores refers to class text.

Positive controls (post hoc). `r6/controls_r6.py` runs SING's AS and IS (class-text target) through the round-6 interventions with the same seeds, with the released translator on all 10k images and with the round-3 ridge translator (refit on the fixed training half with its recorded penalty, scored on the other 5000 images), and `r6/posthoc_r6.py` passes them through the unchanged verdict code. AS, released: estimable on 12 of E12 and 11 of S11, τ_{EQ} 0.39/0.45, median R_B 0.26 (0.25 over E12), mixed. AS, retrained: 0.85/0.85, 0.67 (0.70), mixed. IS, released: 0.12/0.24, 0.57, energy-driven; retrained: 0.12/0.24, 0.79, energy-driven. Both IS controls are therefore flagged, and both AS controls are not. The NuSA share is MIXED (Table 25). ρ is equal on every model after E1, so τ_{EQ} is undefined and the conjunctive rule returns mixed. Under independence of the orders before and after E1, $P(\tau_{\mathrm{EQ}}<0.3)$ is 0.90 for 12 models and 0.89 for 11 (20 000 permutations), so the τ part of the rule says little by itself.

Built-in normalisation (post hoc). ViM's α is refitted after each intervention and scales inversely with the null norm, so the refit score is close to invariant to E1. With α frozen at the real-data value of each fit half, E1 moves the score from 0.876 to 2.5×10^{-4} (ResNet50), 0.843 to 8.5×10^{-5} (DINOv2-L) and 0.876 to 3.1×10^{-5} (VGG16); refitting multiplies α by 4.9, 6.4 and 8.5 on these models (0.8–8.5 over the 13). Frozen-α ViM gives τ_{EQ} $-0.52/-0.49$ and median R_B 1.33 (energy-driven), and the share of its range left after E1 is 1.93. CORE on Πf and the cosine parts of ActSub and CORE do not depend on the scale of the null part. The share of the cross-model range left after E1 (E12, estimable models) is 5.4×10^{-3} (NuSA share), 0.11 and 0.15 (AS, released and retrained), 0.11 (IS, released), 0.29 ($\mathcal{V}$-information), 0.78 (ActSub), 0.90 (ViM), 0.92 (CORE), 1.00 (CORE on Πf) and 0.75 (ViM score s).

$\mathcal{V}$-information (post hoc). Its range across the 13 models is 0.0600 nats, and DenseNet121 alone contributes 86% of it. DenseNet121's null part is tiny ($\rho=1.8\times10^{-8}$) and is scaled up by up to 7×10^{3} in the sweep. Without DenseNet121 the range is 0.0123. Re-applying the estimability rule to this range leaves 7 of E12 and 7 of S11 estimable, against the 9 and 8 that a verdict needs (verdict possible: no). The median R_B over the 12 models is then 0.94, and τ_{EQ} on the estimable models is 0.52/0.52. Of the 13 gains, four are zero to four decimals or negative (VGG16 1.1×10^{-5}, BiT-R50x1 -0.0032, ResNeXt101 -0.0086, DINOv2-L -0.0053), and 4 are within 2 bootstrap SDs of zero. The penalty chosen by inner cross-validation was the largest grid value in 15 of 26 fits, so the probe is heavily shrunk. An independent rerun on ResNet50 gave 0.00103 instead of the audit's 0.00158, which suggested that the chosen penalty flips between runs. We could not reproduce a flip: in 2 reruns with the registered inner cross-validation seed, the penalty chosen for the two halves was $\lambda_A/\lambda_B=1/0.1$ (relative to the default), as in the audit, and the value was 0.00158. The value does depend on the penalty. Over the four combinations of $\{0.1, 1\}$ for the two halves it is 0.00098, 0.00087, 0.00158 and 0.00147 (A/B = 0.1/0.1, 0.1/1, 1/0.1, 1/1), a range of almost a factor of two that contains the value of the independent rerun, and the cross-validated gains that pick λ_B differ by less than 10^{-3} nats (`r6/vinfo_repro_r6.py`). A different BLAS or thread count could therefore change the choice, and we treat the ResNet50 value as known only to this range.

Cross-model and synthetic worlds (descriptive). Table 26 gives τ with SING's order, τ with ρ and the partial association given ρ. None of the four primary scores tracks SING's order: ViM and $\mathcal{V}$-information do not, and ActSub and CORE are inconclusive (τ on E12/S11: ViM 0.03/0.05, ActSub 0.36/0.49, CORE 0.33/0.42, $\mathcal{V}$-information $-0.09/-0.20$). The NuSA share does (0.82/0.82). Table 27 gives each score in the corrected calibration worlds of §4.5: the world-C ratio of the content-free excess (pass if at most 1.5 with all values positive) and τ with ρ in world C and with L in worlds M and D. CORE on Πf is exactly invariant to

rescaling the null part (R_B=0), responds to both content tests on all 13 models, and gives τ=0.45 and 0.38 with SING's order (partial given ρ 0.57 and 0.48). It was pre-registered as descriptive, so it is not counted, and we do not interpret it further here. In the worlds, only CORE passes the world-C bar, and on one base (DINOv2-L). ActSub, CORE and $\mathcal{V}$-information fall as ρ grows at fixed content in world C ($\tau_\rho \leq -0.73$) and rise with content in worlds M and D ($\tau_L \geq 0.88$), whereas ViM rises with ρ in world C and moves against L in worlds M and D, and the NuSA share rises with ρ in world C on all three bases. These worlds are semi-synthetic and do not enter the verdicts.

Deviations. Every test changes only Πf, so the logits are unchanged up to 1.3×10^{-5} (DenseNet121, whose null part is scaled up by factors of about $5{\times}10^2$ to $7{\times}10^3$ to reach the E1 and E2 targets). On every other model the change is below 10^{-12}. The pre-registration's 10^{-8} tolerance referred to the synthetic unit tests. No other deviation from `PREREG_R6.md` occurred. Everything in this appendix marked post hoc is listed in `r6/POSTHOC_R6.md`.

Score	estimable E12	estimable S11	τ_{EQ} E12	τ_{EQ} S11	med. R_B	$\lvert C_1\rvert{\geq}2$ / $\lvert\Delta_{\text{arb}}\rvert{\geq}2$	verdict energy	verdict content
ViM	12	11	0.88	0.89	0.06	9 / 5	not energy-driven	partial
ActSub $S_{\rightarrow}$	12	11	0.79	0.78	0.14	11 / 8	mixed	partial
CORE S_{mem}	12	11	0.85	0.82	0.11	11 / 8	mixed	partial
$\mathcal{V}$-info (class)	11	10	0.20	0.33	0.19	11 / 12	mixed	responsive
Descriptive, not counted								
ViM score s	12	11	1.00	1.00	0.03	4 / 6	not energy-driven	partial
CORE on Πf	12	11	1.00	1.00	$1.6{\times}10^{-16}$	13 / 13	not energy-driven	responsive
$\mathcal{V}$-info (correct)	9	9	0.28	0.28	0.13	7 / 0	mixed	partial
NuSA share	12	11	0.27	0.31	0.89	9 / 7	mixed	partial

Table 25: Round 6, pre-registered outcomes (`PREREG_R6.md`). Estimable: models passing the estimability rule in E12 and S11. τ_{EQ}: Kendall τ between the score after equalising null energy and before. Median R_B: range over the energy sweep relative to the range across models. Counts out of 13 models of $|C_1| \geq 2$ (across-image permutation of Πf) and $|\Delta_{\text{arb}}| \geq 2$ (arbitrary class code, α=0.3), in bootstrap SDs. The content labels are registered outcomes that we do not use as readings, because the C2 flag fires for every score and cannot separate a real loss of information from noise (Appendix G). Counting rule: k=0 of n=4; uninformative given the positive controls of Table 3.

Score	τ E12	τ S11	τ all 13	τ with ρ (E12 / S11)	partial given ρ (E12 / S11)
ViM	0.03 $[-0.44, 0.47]$	0.05 $[-0.47, 0.54]$	0.10 $[-0.36, 0.55]$	0.18 / 0.20	-0.34 / -0.25
ActSub $S_{\rightarrow}$	0.36 $[-0.25, 0.83]$	0.49 $[-0.12, 0.92]$	0.21 $[-0.39, 0.72]$	0.33 / 0.42	0.19 / 0.33
CORE S_{mem}	0.33 $[-0.25, 0.80]$	0.42 $[-0.19, 0.88]$	0.23 $[-0.32, 0.71]$	0.18 / 0.20	0.42 / 0.52
$\mathcal{V}$-info (class)	-0.09 $[-0.69, 0.51]$	-0.20 $[-0.79, 0.48]$	-0.24 $[-0.77, 0.35]$	-0.16 / -0.24	0.21 / 0.08
ViM score s	-0.33 $[-0.84, 0.20]$	-0.35 $[-0.91, 0.25]$	-0.31 $[-0.77, 0.22]$	-0.42 / -0.42	0.37 / 0.35
CORE on Πf	0.45 $[-0.03, 0.86]$	0.38 $[-0.18, 0.83]$	0.28 $[-0.26, 0.77]$	0.24 / 0.16	0.57 / 0.48
$\mathcal{V}$-info (correct)	0.17 $[-0.38, 0.82]$	0.17 $[-0.38, 0.82]$	0.16 $[-0.40, 0.71]$	0.00 / 0.00	0.50 / 0.50
NuSA share	0.82 $[0.57, 0.97]$	0.82 $[0.53, 1.00]$	0.85 $[0.64, 1.00]$	0.97 / 0.96	0.76 / 0.80

Table 26: Round 6, descriptive: Kendall τ of each score with SING's order (percentile 95% CI over 2000 model resamples), τ with ρ, and the partial Spearman correlation with SING's order given ρ, over the estimable models.

	DINOv2-L				ResNet50				ResNeXt101			
Score	C ratio	C τ_ρ	M τ_L	D τ_L	C ratio	C τ_ρ	M τ_L	D τ_L	C ratio	C τ_ρ	M τ_L	D τ_L
ViM	∞	1.00	−1.00	−1.00	∞	1.00	−0.12	−1.00	∞	1.00	−0.24	−1.00
ActSub $S_{\to}$	∞	−1.00	0.97	1.00	∞	−1.00	1.00	1.00	∞	−1.00	1.00	1.00
CORE S_{mem}	1.34	−1.00	1.00	1.00	3.57*	−1.00	1.00	1.00	1.85*	−1.00	1.00	1.00
$\mathcal{V}$-info (class)	2.14*	−0.94	0.97	1.00	∞	−0.73	0.88	0.97	∞	−0.94	1.00	1.00
ViM score s	∞	−1.00	−0.91	1.00	∞	−1.00	−0.15	1.00	∞	−1.00	−0.06	1.00
CORE on Πf	39.22*	−1.00	1.00	1.00	8.06*	−1.00	1.00	1.00	21.49*	−1.00	1.00	1.00
$\mathcal{V}$-info (correct)	2.99*	−1.00	−1.00	−0.42	2.78*	−0.97	0.67	0.97	∞	−1.00	−0.82	0.97
NuSA share	∞	1.00	1.00	0.39	∞	1.00	1.00	0.73	∞	1.00	1.00	0.94

Table 27: Round 6, descriptive: scores in the corrected calibration worlds (residuals permuted across images, seed 100500). C ratio: max/min over the 12 world-C rows of the score minus its content-free floor (one across-image permutation); * fails the pre-stated 1.5 bar or has a non-positive excess; ∞ marks a non-positive minimum. τ_ρ: τ with ρ in world C; τ_L: τ with L in worlds M and D.

H Reproducibility and compute

`make reproduce` regenerates every table, figure and number in this paper from committed aggregates in under a minute. This checks consistency, not the feature-level computation. `make reproduce-full` reruns the feature-level analyses from the private ImageNet features: about 2 h wall on 8 CPU cores, most of it the holdout. `make holdout-fast` runs the reduced holdout (36 min on 8 cores). It fails its pre-registered accuracy target on a deterministic quantity, so a faster machine cannot make it pass, and we did not re-time it. The timing itself may be affected by contention: the machine is shared, an earlier attempt was aborted because other jobs were running, the load average was not recorded for the timed run, and the job used about 7.3 of the 8 cores. The round-4 analyses (`r4/`) run on CPU from the same private features (`make reproduce-r4`); extracting the true DINO features needs a GPU. The round-5 analyses (`r5/`, `make reproduce-r5`) take about 20 min for the corrected calibration worlds on 4 cores and 9 min for the organisation-score gates on 4 cores, on a shared machine. The round-6 audit (`r6/`, `make reproduce-r6`) took 3–74 min per model and about 1 h per calibration base with 2 threads per process, four processes in parallel on the shared 8-core machine. The 4096-dimensional VGG models take longest. The post hoc round-6 controls (`make reproduce-r6-posthoc`) took from about a minute per model to about 40 min for each VGG model, and the $\mathcal{V}$-information penalty rerun about 16 min, on 2 threads of a heavily loaded 2-core machine. The post hoc round-5 SO_{null} rerun (`r5/so_exact_r5.py`) takes about 3 min. Feature extraction (13 models on the 10k subset, the Exp 1 features and the five ViT-L/16 candidates on the 50k validation set) needs ImageNet and a GPU. Two jobs were timed: the Exp 1 extraction (1.4 GPU-hours on one cloud GPU) and the re-extraction of the five ViT-L/16 candidates for the translator check (1.9 h on two T4 GPUs, about 3.8 GPU-hours including weight downloads). The other extraction jobs were not timed; we estimate under 10 GPU-hours for all extraction.

Seeds. Family A random draws 10000+r ($r < 20$); family B 50000+r; family C subspaces 30000+j and controls 60000+j; Exp 1 20 seeds per d; E3 plants seed 0, bootstrap 10000× seed 1; model bootstrap 2000× seed 0; claim bootstrap 1000× seed 0; E9 and holdout α seed 1; flatness bases 20000+j; PC-band draws 40000+j; A3 rotations 80000+j; leak-free surrogates 70000+r; translator split permutation seed 0. Round 4: surrogates 92000+r (zero-mean, isotropic, permuted) and 93000+r (within-class permutation); class-mean derangements 94000+r; NCM shuffled labels 95000; selectivity random rotations 96000+j; NZS control 97000; synthetic geometry model 98000; geometry random subspaces 1000d+r; selectivity and geometry image bootstraps seed 0. Round 5: calibration residual permutation 99000; XCC Haar frames 99100+s; SO folds 99200; permutation nulls 99300; world-S plants 99400 and inheritance plant 99401; G1 rotation 99500; XCC self-term folds 99600; class-mean derangements 94000+r. Round 6: halves 100100; ActSub reference subsets 100150/100151; across-image permutations 100200+r; within-class permutations 100250+r; class-code plants 100400; world residual permutation 100500; image bootstrap 100600; $\mathcal{V}$-information inner CV 100700.